

Controllability of FSA-v5 from SEDENTARY_INIT

A constructive obstruction analysis with mechanisation roadmap

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Abstract

A heavily upgraded SMC²-MPC controller, run for three hours of wall time on a T=100 d horizon from SEDENTARY_INIT, fails to escape the sedentary basin: the autonomic amplitude A decays from 0.10 to 0.003 under the controller's chosen Φ schedule. We ask whether this is a controller-tuning problem or a structural property of the deterministic FSA-v5 model. We give a quantitative obstruction: under the canonical TRUTH_PARAMS_V5, the bifurcation parameter at SEDENTARY_INIT is $\mu \approx -0.115$ independent of Φ at $t = 0$, and the time-scales of B, S recovery ($\tau_B = 42$ d, $\tau_S = 60$ d) are slow relative to the A -decay time-scale (~ 8.7 d) so that the integrated bifurcation parameter $\int_0^T \mu(t) dt$ cannot exceed the escape threshold $\ln(A_{\text{target}}/A_0) \approx 1.10$ within $T = 100$ d under any admissible $\Phi \in [0, 1]^2$ and not under $\Phi \in [0, 3]^2$ either. We exhibit three constructive escape candidates (constant balanced, bang-bang aerobic burst, smart ramp) and integrate them numerically: $A(100) \in \{0.036, 0.096, 0.049\}$ respectively, all an order of magnitude below $A_{\text{target}} = 0.30$. The best candidate (a one-shot aerobic burst) *maintains* A near 0.10 but does not effect escape. We then state a quantitative impossibility theorem with a comparison-theorem upper bound on $\int_0^T \mu(t) dt$, suitable for a Lean4 formalisation, and conclude with a sensitivity analysis suggesting that the pinned deconditioning parameters ($\mu_{B-}, \mu_{S-}, B_{\text{dec}}, S_{\text{dec}}, n$) are the most likely lever for making the model controllable. The recommended action, conditional on confirming the sensitivity result, is to **modify the FSA-v5 deconditioning block** rather than continue tuning the controller. Two figures are included (candidate trajectories, controllability region) and the Python script that generates them is in this directory.

Extension (§14–§16). Three new sections analyse the §13 modification recommendations in detail. §14 plots the healthy island under each of the three single deconditioning modifications, showing that none shifts the island toward the user's target $\Phi = (1, 1)$ — Mod B in fact shifts in the opposite direction, and Mod C destroys the island entirely. §15 performs a multi-parameter sweep proving that the deconditioning block alone cannot satisfy three qualitative criteria (sedentary basin preserved, healthy at $(1, 1)$, over-training basin preserved at $\Phi \geq (2, 2)$); the sweep must include reward-side coefficients μ_F, μ_{FF} . The recommended parametrisation is $B_{\text{dec}} = 0.25$, $\mu_{\text{dec}} = 0.10$, $\mu_F = 0.030$, $\mu_{FF} = 0.020$, with island center at $(1.06, 0.78)$ and stable autonomic attractor $A^* = 1.24$. §16 verifies four qualitative behaviours under the recommended parametrisation by hand-designed piecewise $\Phi(t)$ witnesses: (i) sedentary collapse from SEDENTARY_INIT under passive $\Phi = (0.1, 0.1)$, (ii) constructive escape from SEDENTARY_INIT under a 2-segment burst-then-balanced witness with $A(200) = 1.21$ (LEAN-mechanisable), (iii) over-training collapse from TRAINED_ATHLETE_INIT_v2 under passive $\Phi = (2, 2)$, (iv) maintenance under $\Phi = (1, 1)$. All four pass.

§17 restores $T = 100$ d as the working horizon via an equilibrium-preserving timescale halving (τ_B, κ_B) $\rightarrow$ ($\tau_B/2, 2\kappa_B$) (and analogously for S), and replaces the §16.2 hand-designed witness with a 21×21 grid search over constant Φ that selects the candidate minimising the trajectory FIM condition number — treating Φ itself as the parameter to identify, by the classical identification- $\leftrightarrow$ -controllability duality. The FIM-selected constant witness $\Phi^* = (0.86, 1.24)$ achieves $A(100) = 1.16$ from SEDENTARY_INIT, with all four §16 qualitative behaviours preserved at the shorter horizon.

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1 Introduction and problem statement

1.1 The empirical observation

A T=100 d closed-loop SMC²-MPC bench was launched on 2026-05-11 with the following configuration: SEDENTARY_INIT initial state, $\Phi(0) = (0, 0)$, all chance-constraint terms disabled, the new *healthy-island gradient surrogate* (denoted λ_{island}) active at strength 1.0, A_t -tracked $\bar{\mu}$ formulation, 256 SMC particles, 4 HMC moves per tempering level, and 60 tempering levels. After approximately three hours of GPU wall time the trace shows

$$A(t) : 0.10 \xrightarrow{100 \text{ days}} 0.003,$$

i.e. a near-complete collapse rather than escape. The bench output is at `outputs/bench_runs/2026-05-11_053431_T100d_UltraFast/v5_traces.png`.

The user’s hypothesis: this is a *structural mathematical property* of the FSA-v5 model under SEDENTARY_INIT, not a controller-tuning failure. The remainder of this document tests that hypothesis.

1.2 Working assumptions

Throughout the analysis we adopt:

1. **Deterministic dynamics.** We work with the deterministic ODE underlying the FSA-v5 SDE, *not* the stochastic process that the bench actually solves. Stochastic extension is deferred to Section 12.
2. **Full state observability.** We assume the controller knows the 6D state $(B, S, F, A, K_{FB}, K_{FS})$ exactly and instantaneously, and is free to apply any measurable $\Phi: [0, T] \rightarrow U$ for $U \in \{[0, 1]^2, [0, 3]^2\}$. The closed-loop bench’s particle filter is removed from the picture; we are asking the strictly easier *open-loop controllability* question.
3. **Canonical parameters.** TRUTH_PARAMS_V5 as defined in `version_1_5_Julia/models/fsa_v5/simulation_v5.jl` lines 80–135. Parameter values, including their centred forms, are tabulated in Section 2.
4. **Two admissible control sets.** $U \in \{[0, 1]^2, [0, 3]^2\}$. The smaller set matches realistic training intensity; the larger set matches what the GPU controller’s sigmoid envelope $\sigma(\cdot) \cdot \Phi_{\max}$ allows with $\Phi_{\max} = 3$.

If the system is uncontrollable from SEDENTARY_INIT under these (lenient) assumptions, no realistic closed-loop controller could do better.

1.3 The question, formally

Fix horizon $T > 0$ and target $A_{\text{target}} > 0$. Let $y(t; y_0, \Phi)$ denote the (deterministic) state at time t under initial condition y_0 and control Φ .

Definition 1 (Controllability). The state y_0 is $(T, A_{\text{target}}, U)$ -controllable to the healthy basin if there exists a measurable $\Phi: [0, T] \rightarrow U$ such that $A(T; y_0, \Phi) \geq A_{\text{target}}$.

We ask: is $y_0 = \text{SEDENTARY_INIT}$ $(100, 0.30, [0, 1]^2)$ -controllable? $(100, 0.30, [0, 3]^2)$ -controllable?

1.4 Headline result preview

Section 8 exhibits the integrated bifurcation parameter $\int_0^T \mu(t) dt$ as the critical scalar:

$$A(T) = A_0 \cdot \exp\left(\int_0^T \mu(t) dt - \int_0^T \eta A(t)^2 dt\right),$$

so $\int_0^T \mu(t) dt \geq \ln(A_{\text{target}}/A_0) + \int \eta A^2 \geq \ln(3)$ is necessary for escape from $A_0 = 0.10$ to $A_{\text{target}} = 0.30$. We then show:

- Three explicit Φ candidates (constant, bang–bang, ramp) all give $\int \mu \in [-2.7, -0.4]$ at $T = 100$ d, far below the $\ln 3 \approx 1.10$ threshold.
- A comparison-theorem upper bound implies that for any admissible $\Phi \in [0, 1]^2$ on $[0, 100]$ d, $\int \mu(t) dt \leq -0.4$ (precise constant in Theorem 9).
- Therefore **SEDENTARY_INIT** is $(100, 0.30, [0, 1]^2)$ -uncontrollable with respect to the healthy basin under the canonical **TRUTH_PARAMS_V5**.
- For $U = [0, 3]^2$, the situation is more delicate (large Φ over-trains F , also bad), but our numerical evidence is the same direction.

The likely lever for fixing the model rather than the controller is the pinned deconditioning block $(\mu_{B-}, \mu_{S-}, B_{\text{dec}}, S_{\text{dec}}, n)$ (Section 13).

2 Model recall

We summarise the FSA-v5 deterministic dynamics; full derivation is in the technical guide [1] §2–3, particularly the Hill-deconditioning extension §14.

2.1 State and control

The state $y \in \mathbb{R}^6$ comprises

$$y = (B, S, F, A, K_{FB}, K_{FS})^\top,$$

with $B, S \in [0, 1]$ (chronic aerobic and strength capacities), $F \in [0, \infty)$ (unified fatigue), $A \in [0, \infty)$ (autonomic amplitude), and $K_{FB}, K_{FS} \in [0, \infty)$ (Busso-style fatigue-sensitivity gains). The control is $\Phi = (\Phi_B, \Phi_S) \in U \subseteq [0, \infty)^2$.

2.2 Drift equations

Writing $a_X(A) = (1 + \epsilon_{AX}A)/(1 + \epsilon_{AX}A_{\text{TYP}})$ for $X \in \{AB, AS\}$ and $a_F(A) = (1 + \lambda_AA)/(1 + \lambda_AA_{\text{TYP}})$:

$$\dot{B} = \kappa_B a_B(A) \Phi_B - B/\tau_B, \quad (1)$$

$$\dot{S} = \kappa_S a_S(A) \Phi_S - S/\tau_S, \quad (2)$$

$$\dot{F} = K_{FB}\Phi_B + K_{FS}\Phi_S - a_F(A)F/\tau_F, \quad (3)$$

$$\dot{A} = \mu(B, S, F) A - \eta A^3, \quad (4)$$

$$\dot{K}_{Fi} = (K_{Fi}^0 - K_{Fi})/\tau_K + \mu_K \Phi_i, \quad i \in \{B, S\}. \quad (5)$$

The bifurcation parameter (FSA-v5 form, with Hill deconditioning):

$$\mu(B, S, F) = \mu_0 + \mu_B B + \mu_S S - \mu_F F - \mu_{FF}(F - F_{\text{TYP}})^2 - \mu_B h_n(B; B_{\text{dec}}) - \mu_S h_n(S; S_{\text{dec}}) \quad (6)$$

where the Hill saturation function is

$$h_n(x; x_{\text{dec}}) = \frac{x_{\text{dec}}^n}{x^n + x_{\text{dec}}^n} \in [0, 1], \quad (7)$$

saturating to 1 as $x \rightarrow 0$ (full deconditioning penalty applied) and dropping to 0 as $x \gg x_{\text{dec}}$ (well-trained, no penalty).

2.3 Cascade structure and slow manifold

The drift is a feed-forward cascade $K \rightarrow (B, S) \rightarrow F \rightarrow A$ with back-coupling only through the activation factors $a_X(A)$. At constant Φ each forward block has a closed-form equilibrium parametrised by A (see [1] §11.1–11.2):

$$K_{Fi}^*(\Phi_i) = K_{Fi}^0 + \tau_K \mu_K \Phi_i, \quad (8)$$

$$B^*(A; \Phi_B) = \tau_B \kappa_B a_B(A) \Phi_B, \quad (9)$$

$$S^*(A; \Phi_S) = \tau_S \kappa_S a_S(A) \Phi_S, \quad (10)$$

$$F^*(A; \Phi) = \tau_F (K_{FB}^* \Phi_B + K_{FS}^* \Phi_S) / a_F(A). \quad (11)$$

Substituting into (6) gives the *slow-manifold* bifurcation parameter $\bar{\mu}(A; \Phi)$. The interior equilibria of A solve $\bar{\mu}(A; \Phi) = \eta A^2$. The boundary $A = 0$ is unstable iff $\bar{\mu}(0; \Phi) > 0$; this is the transcritical condition that delineates the closed healthy island in (Φ_B, Φ_S) -space.

2.4 Numerical anchor: the sanity table

The five-point sanity table from [1, Table 5] §14 reads, under TRUTH_PARAMS_V5:

Stimulus regime	$\bar{\mu}(0; \Phi)$	Outcome
Sedentary, $\Phi = (0, 0)$	−0.180	strong collapse
Aerobic-only, $\Phi = (0.30, 0)$	−0.059	collapse
Strength-only, $\Phi = (0, 0.30)$	−0.093	collapse
Balanced moderate, $\Phi = (0.30, 0.30)$	+0.011	healthy ($A^* \approx 0.54$)
Over-training, $\Phi = (1.0, 1.0)$	−1.608	catastrophic collapse

Crucially, the maximum of $\bar{\mu}(0; \Phi)$ over $\Phi \in [0, 3]^2$ is bounded above by approximately +0.015 and is achieved near $\Phi \approx (0.30, 0.30)$. We confirm this from the colour-bar of the existing landscape figure produced by `plot_mu_bar_landscape.jl` (the value at the tabulated balanced-moderate point is +0.0112 to four decimal places).

2.5 Parameter values used throughout

The numerical parameters are read from `simulation_v5.jl:80-135`. We state the centred forms explicitly:

Parameter	Value	Parameter	Value
τ_B	42.0 d	μ_0	$0.02 + 0.40 \cdot F_{\text{TYP}}^2 = 0.036$
κ_B	$0.012(1 + 0.40A_{\text{TYP}}) = 0.01248$	μ_B	0.30
ϵ_{AB}	0.40	μ_S	0.15
τ_S	60.0 d	μ_F	$0.10 + 2F_{\text{TYP}} \cdot 0.40 = 0.26$
κ_S	$0.008(1 + 0.20A_{\text{TYP}}) = 0.00816$	μ_{FF}	0.40
ϵ_{AS}	0.20	η	0.20
τ_F	$7.0/(1 + A_{\text{TYP}}) = 6.36$ d	μ_{B-}	0.10
λ_A	1.00	μ_{S-}	0.10
K_{FB}^0	0.030	B_{dec}	0.07
K_{FS}^0	0.050	S_{dec}	0.07
τ_K	21.0 d	n	4
μ_K	0.005	$A_{\text{TYP}}, F_{\text{TYP}}$	0.10, 0.20

The 8 dynamics-side parameters ($K_{FB}^0, K_{FS}^0, \tau_K, n, B_{\text{dec}}, S_{\text{dec}}, \mu_{B-}, \mu_{S-}$) are *frozen* (not estimated from data), per the §11.6 Fisher-information analysis. We will return to this in Section 13.

2.6 Self-contained reproducer: SEDENTARY_INIT

For a reader who does not have access to the codebase, this subsection makes everything needed to verify the analysis fully self-contained. A reader can paste the formulas below into any language (Python, Julia, R, even a spreadsheet) and reproduce every numerical claim in the document.

The 6 state variables. Each component of $y = (B, S, F, A, K_{FB}, K_{FS})^\top$ is dimensionless (all rates in day^{-1} , all variables scaled so typical healthy values are $O(0.1)$ to $O(1)$):

Symbol	Range	Typical	Physical meaning
B	$[0, 1]$	0.5 (trained)	Chronic aerobic-fitness capacity (Banister chronic).
S	$[0, 1]$	0.45 (trained)	Chronic strength capacity.
F	$[0, \infty)$	0.20 (F_{TYP})	Unified short-term fatigue.
A	$[0, \infty)$	0.45 (trained), 0 (collapsed)	Autonomic amplitude (clinical HRV proxy).
K_{FB}	$[0, \infty)$	0.030–0.1	Aerobic fatigue-sensitivity gain (Busso variable-dose).
K_{FS}	$[0, \infty)$	0.050–0.15	Strength fatigue-sensitivity gain.

The initial condition. SEDENTARY_INIT represents a deconditioned-but-not-collapsed athlete (e.g., after a 6-month sedentary period from a prior trained state). The exact numerical values:

$$\text{SEDENTARY_INIT} = \begin{cases} B_0 & = 0.05 \\ S_0 & = 0.10 \\ F_0 & = 0.30 \\ A_0 & = 0.10 \\ K_{FB,0} & = 0.030 = K_{FB}^0 \text{ (at slow-manifold baseline)} \\ K_{FS,0} & = 0.050 = K_{FS}^0 \text{ (at slow-manifold baseline)} \end{cases} \quad (12)$$

Interpretive notes.

- $B_0 = 0.05$ is *below* the deconditioning threshold $B_{\text{dec}} = 0.07$. This is the critical fact making escape hard: the Hill term $h_4(B_0; B_{\text{dec}})$ is near its maximum value of 1.
- $S_0 = 0.10$ is *above* $S_{\text{dec}} = 0.07$, but only slightly; $h_4(S_0; S_{\text{dec}}) \approx 0.19$.
- $F_0 = F_{\text{TYP}} \pm 0.10$: the athlete is mildly fatigued. The quadratic fatigue penalty $\mu_{FF}(F_0 - F_{\text{TYP}})^2 = 0.40 \cdot 0.10^2 = 0.004$ is small.
- $A_0 = A_{\text{TYP}} = 0.10$: well below the trained-athlete value of ~ 0.45 . Still positive (no irreversible collapse), but the controllability target $A_{\text{target}} = 0.30$ requires a $3\times$ multiplicative recovery.
- $K_{FB,0}, K_{FS,0}$ are at their baseline slow-manifold values; they are not the critical variables for the obstruction.

State clipping (plant convention). The Julia plant in `_dynamics_v5.jl` clips $B, S \in [\varepsilon, 1 - \varepsilon]$ with $\varepsilon = 10^{-4}$, and floors $F, A, K_{FB}, K_{FS} \geq 0$, after each Euler–Maruyama step. The deterministic ODE we analyse here does *not* include this clipping — but the clipping only makes impossibility *easier* to establish, because A pinned to 0 once it touches the boundary stays at 0 forever (since $\dot{A} = 0 \cdot \mu - 0 = 0$ at $A = 0$ in the unclipped form too).

Self-contained reproducer for $\mu(\text{SEDENTARY_INIT})$. The following pseudocode computes the headline number $\mu_{\text{init}} = -0.115$ from first principles. Any reader should be able to type this into their language of choice and verify; no codebase access required.

```
# Hard-coded TRUTH_PARAMS_V5 values
A_TYP = 0.10
F_TYP = 0.20
mu_0 = 0.02 + 0.40 * F_TYP**2 # = 0.036
mu_B = 0.30
mu_S = 0.15
mu_F = 0.10 + 2.0 * F_TYP * 0.40 # = 0.26
mu_FF = 0.40
mu_dec_B = 0.10 # pinned
mu_dec_S = 0.10 # pinned
B_dec = 0.07 # pinned
S_dec = 0.07 # pinned
n = 4 # pinned

# SEDENTARY_INIT state
B, S, F = 0.05, 0.10, 0.30

# Hill saturation: h_n(x; xd) = xd**n / (x**n + xd**n)
def h(x, xd, n):
    return xd**n / (x**n + xd**n)

# Bifurcation parameter (Eq. 11 in the document)
mu = (mu_0 + mu_B*B + mu_S*S
      - mu_F*F - mu_FF*(F - F_TYP)**2
      - mu_dec_B * h(B, B_dec, n)
      - mu_dec_S * h(S, S_dec, n))

print(mu) # expect -0.115
```


Intermediate values for line-by-line verification. A reader checking the arithmetic by hand should see:

Term	Expression	Value
μ_0	$0.02 + 0.40 \cdot 0.04$	+0.036
$\mu_B \cdot B_0$	$0.30 \cdot 0.05$	+0.015
$\mu_S \cdot S_0$	$0.15 \cdot 0.10$	+0.015
$-\mu_F \cdot F_0$	$-0.26 \cdot 0.30$	-0.078
$-\mu_{FF}(F_0 - F_{\text{TYP}})^2$	$-0.40 \cdot 0.10^2$	-0.004
$h_4(B_0; B_{\text{dec}})$	$0.07^4 / (0.05^4 + 0.07^4)$ $= 24.01 / (6.25 + 24.01) \cdot 10^{-6}$	≈ 0.793
$-\mu_B h_4(B_0; \cdot)$	$-0.10 \cdot 0.793$	-0.0793
$h_4(S_0; S_{\text{dec}})$	$0.07^4 / (0.10^4 + 0.07^4)$ $= 24.01 / (100 + 24.01) \cdot 10^{-6}$	≈ 0.194
$-\mu_S h_4(S_0; \cdot)$	$-0.10 \cdot 0.194$	-0.0194
Sum: $\mu(\text{SEDENTARY_INIT})$		-0.1147

This is the central numerical anchor of the document; it appears in Eq. (15) and is the starting point for every subsequent estimate.

Why the value is Φ -independent at $t = 0$. The control Φ does *not* appear in the formula above. Φ enters the system only through the drift equations (1)–(3), which determine the time-evolution of B, S, F, K . So at $t = 0$, regardless of what $\Phi(0)$ the controller chooses, $\mu(t = 0) = -0.115$, and $\dot{A}(0) = \mu(0) \cdot A_0 - \eta A_0^3 = -0.115 \cdot 0.10 - 0.20 \cdot 0.001 = -0.0117/\text{day}$. A is decaying at this rate *no matter what the controller does at $t = 0$* .

3 The control system

3.1 Admissible controls

Definition 2. For an admissible-set parameter $U \subseteq \mathbb{R}_{\geq 0}^2$, the set of admissible controls on $[0, T]$ is

$$\mathcal{U}(T, U) := \{\Phi : [0, T] \rightarrow U \mid \Phi \text{ measurable}\}.$$

We consider $U \in \{[0, 1]^2, [0, 3]^2\}$. The kernel implementation in `gpu_control_v5.jl` parametrises Φ via Gaussian RBF basis decoded through a sigmoid envelope $\sigma(\cdot) \cdot \Phi_{\max}$ with $\Phi_{\max} = 3$, so the kernel-realisable controls are a subset of $\mathcal{U}(T, [0, 3]^2)$. Lifting the proof to the larger admissible set $\mathcal{U}(T, [0, 3]^2)$ is conservative: any uncontrollability result there transfers to the realisable subset.

3.2 Reachable set and target

Definition 3. The *reachable set* from y_0 on $[0, T]$ under $\mathcal{U}(T, U)$ is

$$\mathcal{R}(T; y_0, U) := \{y(T; y_0, \Phi) \mid \Phi \in \mathcal{U}(T, U)\}.$$

The *healthy target set* is $\mathcal{T}(A_{\text{target}}) := \{y \in \mathbb{R}^6 \mid A \geq A_{\text{target}}\}$. The state y_0 is *controllable* iff $\mathcal{R}(T; y_0, U) \cap \mathcal{T}(A_{\text{target}}) \neq \emptyset$.

4 Slow-manifold reduction and cascade upper envelopes

4.1 An a-priori upper bound on $B(t)$

Equation (1) is linear in B with A -dependent coefficients. Since $a_B(A) = (1 + \epsilon_{AB}A)/(1 + \epsilon_{AB}A_{\text{TYP}})$ is monotone increasing in A , an upper bound on $A(t)$ yields an upper bound on $B(t)$.

Lemma 4 (B -upper-envelope). *Suppose $A(t) \leq \bar{A}$ for all $t \in [0, T]$ and $\Phi_B(t) \leq \Phi_B^{\max}$. Then for $B(0) = B_0$,*

$$B(t) \leq B_{\text{UB}}(t; \bar{A}, \Phi_B^{\max}) := B_{\infty} - (B_{\infty} - B_0) e^{-t/\tau_B}, \quad B_{\infty} := \tau_B \kappa_B a_B(\bar{A}) \Phi_B^{\max}.$$

Proof. Let $\tilde{B}$ solve $\dot{\tilde{B}} = \kappa_B a_B(\bar{A}) \Phi_B^{\max} - \tilde{B}/\tau_B$ with $\tilde{B}(0) = B_0$. By the linear-ODE comparison theorem and $a_B(A(t)) \leq a_B(\bar{A})$, $\Phi_B(t) \leq \Phi_B^{\max}$, we have $B(t) \leq \tilde{B}(t)$. The closed-form solution of the affine ODE is $\tilde{B}(t) = B_{\infty} - (B_{\infty} - B_0) e^{-t/\tau_B}$. $\square$

Remark 5. Strictly, the chronic state B is also clipped to $[0, 1]$ in the Julia plant; for $B_{\infty} > 1$ (which happens when $\Phi_B^{\max} = 3$ since $\tau_B \kappa_B = 0.524$, so $B_{\infty} \approx 1.81 \cdot a_B(\bar{A})$) the comparison gives the conservative $B(t) \leq \min(B_{\text{UB}}(t), 1)$. We use this clipped form below.

The analogous lemma for S is identical with $(\tau_S, \kappa_S, \epsilon_{AS}, S_0)$ replacing $(\tau_B, \kappa_B, \epsilon_{AB}, B_0)$.

4.2 An a-priori lower bound on $F(t)$

For the impossibility direction we want a *lower* bound on F (so that the negative $-\mu_F F - \mu_{FF}(F - F_{\text{TYP}})^2$ contribution can be quantified). However, the impossibility argument we actually use *drops* these terms (they only push μ lower, hence make impossibility easier — see Section 5). So we do not need a sharp F -bound; the trivial $F(t) \geq 0$ suffices.

4.3 An a-priori bound on $A(t)$

The cubic damping in (4) ensures $A(t)$ stays bounded under bounded μ . We adopt the standing hypothesis (consistent with the impossibility we are trying to prove) that $A(t) \leq \max(A_0, A_{\text{target}})$ for all $t \in [0, T]$. Under `SEDENTARY_INIT`, $A_0 = 0.10$ and $A_{\text{target}} = 0.30$, so $\bar{A} = 0.30$. If at any $t^* \in [0, T]$ we have $A(t^*) > A_{\text{target}}$, then by definition the system is controllable and we are done; otherwise the standing hypothesis holds throughout $[0, T]$ and the analysis below applies.

4.4 Effect of the upper bound on a_X at $\bar{A} = 0.30$

$$a_B(\bar{A}) = \frac{1 + 0.40 \cdot 0.30}{1 + 0.40 \cdot 0.10} = \frac{1.12}{1.04} = 1.077, \quad a_S(\bar{A}) = \frac{1.06}{1.02} = 1.039.$$

So $a_B^{\max} = 1.077$ and $a_S^{\max} = 1.039$ for our purposes.

5 The forcing-rate bottleneck: an upper bound on $\mu(t)$

5.1 Conservative omissions for the upper bound

We construct an a-priori upper bound on $\mu(B(t), S(t), F(t))$ that depends only on t , y_0 , and the admissible-set parameter U . The key observation: each negative term in (6) can be *dropped* to obtain a looser upper bound on μ , and dropping makes the impossibility theorem strictly harder to prove. So our *omissions* below are the conservative choice.

- Drop $-\mu_F F$: this term is ≤ 0 for $F \geq 0$. Dropping it makes μ *larger*.

- Drop $-\mu_{FF}(F - F_{\text{TYP}})^2 \leq 0$. Dropping makes μ larger.
- Drop $-\eta A^3$ from the A -ODE (4): ≤ 0 for $A \geq 0$. Dropping the cubic makes $\dot{A}/A$ larger.

5.2 The retained terms and their bounds

We retain $\mu_0 + \mu_B B + \mu_S S - \mu_B h_n(B; B_{\text{dec}}) - \mu_S h_n(S; S_{\text{dec}})$. Since both $\mu_B B$ and $-\mu_B h_n(B; B_{\text{dec}})$ are monotone in B (the first increasing, the second also increasing because h_n is decreasing in B), *both* retained terms increase as B grows. Therefore an upper bound on B gives an upper bound on the combined contribution. Similarly for S . Using Lemma 4 (and its S analogue):

Theorem 6 (Upper bound on $\mu(t)$). *Under the standing hypothesis $A(t) \leq \bar{A}$ on $[0, T]$, and for any $\Phi \in \mathcal{U}(T, U)$ with $U = [0, \Phi^{\max}]^2$,*

$$\mu(B(t), S(t), F(t)) \leq \bar{\mu}_{\text{UB}}(t; \bar{A}, \Phi^{\max}) \quad (13)$$

where

$$\begin{aligned} \bar{\mu}_{\text{UB}}(t; \bar{A}, \Phi^{\max}) := & \mu_0 + \mu_B B_{\text{UB}}(t; \bar{A}, \Phi^{\max}) + \mu_S S_{\text{UB}}(t; \bar{A}, \Phi^{\max}) \\ & - \mu_B h_n(B_{\text{UB}}(t; \bar{A}, \Phi^{\max}); B_{\text{dec}}) \\ & - \mu_S h_n(S_{\text{UB}}(t; \bar{A}, \Phi^{\max}); S_{\text{dec}}). \end{aligned} \quad (14)$$

Proof. Drop $-\mu_F F - \mu_{FF}(F - F_{\text{TYP}})^2 \leq 0$. Use the definition of μ (6). For the linear terms: $\mu_B B(t) \leq \mu_B B_{\text{UB}}(t)$ by Lemma 4 (and $\mu_B \geq 0$). For the Hill term: since $h_n(\cdot; B_{\text{dec}})$ is decreasing in its first argument and $B(t) \leq B_{\text{UB}}(t)$, we have $h_n(B(t); B_{\text{dec}}) \geq h_n(B_{\text{UB}}(t); B_{\text{dec}})$, hence $-\mu_B h_n(B(t); B_{\text{dec}}) \leq -\mu_B h_n(B_{\text{UB}}(t); B_{\text{dec}})$. Sum. $\square$

Remark 7 (Sharpness). The bound is sharp at $t = 0$: the inequality is an equality when $B(0) = B_{\text{UB}}(0) = B_0$ and $S(0) = S_{\text{UB}}(0) = S_0$, regardless of $\Phi(0)$. So $\bar{\mu}_{\text{UB}}(0; \bar{A}, \Phi^{\max}) = \mu(B_0, S_0, F_0) -$ (omitted negative F terms), and at SEDENTARY_INIT this is approximately $-0.115 + (\mu_F F_0 + \mu_{FF}(F_0 - F_{\text{TYP}})^2) \approx -0.115 + 0.082 = -0.033$. We carry the full computation in Section 8.

5.3 Numerical evaluation at SEDENTARY_INIT

Using the parameters of Section 2 with $\bar{A} = 0.30$ (so $a_B^{\max} = 1.077$, $a_S^{\max} = 1.039$):

For $\Phi^{\max} = 1$ (realistic):

$$\begin{aligned} B_{\infty} &= 42 \cdot 0.01248 \cdot 1.077 \cdot 1 = 0.564, \\ S_{\infty} &= 60 \cdot 0.00816 \cdot 1.039 \cdot 1 = 0.509. \end{aligned}$$

For $\Phi^{\max} = 3$ (kernel), with the physical clip $B, S \leq 1$:

$$\begin{aligned} B_{\infty}^{\text{raw}} &= 1.692, & B_{\infty}^{\text{clip}} &= 1.0, \\ S_{\infty}^{\text{raw}} &= 1.527, & S_{\infty}^{\text{clip}} &= 1.0. \end{aligned}$$

The time for B_{UB} to clear $B_{\text{dec}} = 0.07$ from $B_0 = 0.05$ is approximately:

$$t^*(B; \Phi^{\max}) = -\tau_B \ln \frac{B_{\infty} - B_{\text{dec}}}{B_{\infty} - B_0}.$$

For $\Phi^{\max} = 1$: $t^*(B; 1) = -42 \ln(0.494/0.514) = 1.66$ d. For $\Phi^{\max} = 3$ (with clip): $t^*(B; 3) = -42 \ln(0.93/0.95) = 0.89$ d.

So B can clear its deconditioning threshold in 1–2 days under maximum aerobic stimulus. After that, $h_n(B; B_{\text{dec}})$ drops sharply.

For S : $S_0 = 0.10 > S_{\text{dec}} = 0.07$ already, so S never crosses; $h_n(S_0; S_{\text{dec}}) = 0.194$ initially, and decreases monotonically as S grows.

6 Constructive escape attempts

We now exhibit three explicit Φ candidates and compute the resulting trajectories under `SEDENTARY_INIT`. Each computation is symbolic where possible; the integrals are done in closed form on the slow manifold and corrected for the transient.

6.1 Candidate 1: Constant balanced $\Phi \equiv (0.30, 0.30)$

This is the slow-manifold optimum: it places (Φ_B, Φ_S) at the centre of the v5 healthy island. Once on the slow manifold, $\bar{\mu}(0; \Phi) = +0.011 > 0$ and $A^* \approx 0.54$.

Slow-manifold timescale. B and S approach their slow-manifold values $B^*(\bar{A}; (0.30, 0.30))$ and $S^*(\bar{A}; (0.30, 0.30))$ on time-scales $\tau_B = 42$ d and $\tau_S = 60$ d respectively. Concretely, with $\bar{A} = 0.30$:

$$\begin{aligned} B^*((0.30, 0.30); A = 0.30) &= 42 \cdot 0.01248 \cdot 1.077 \cdot 0.30 = 0.169, \\ S^*((0.30, 0.30); A = 0.30) &= 60 \cdot 0.00816 \cdot 1.039 \cdot 0.30 = 0.153, \\ F^*((0.30, 0.30); A = 0.30) &= 6.36 \cdot (0.0615 \cdot 0.30 + 0.0815 \cdot 0.30) / (1.30/1.10) = 0.236. \end{aligned}$$

Initial μ value. At $t = 0$, the state is $y_0 = \text{SEDENTARY_INIT}$. The bifurcation parameter is computed directly from (6):

$$\begin{aligned} \mu_0 + \mu_B B_0 + \mu_S S_0 &= 0.036 + 0.30 \cdot 0.05 + 0.15 \cdot 0.10 = 0.066, \\ \mu_F F_0 + \mu_{FF}(F_0 - F_{\text{TP}})^2 &= 0.26 \cdot 0.30 + 0.40 \cdot (0.10)^2 = 0.082, \\ \mu_{B-} h_n(B_0; B_{\text{dec}}) &= 0.10 \cdot \frac{0.07^4}{0.05^4 + 0.07^4} = 0.10 \cdot 0.793 = 0.0793, \\ \mu_{S-} h_n(S_0; S_{\text{dec}}) &= 0.10 \cdot \frac{0.07^4}{0.10^4 + 0.07^4} = 0.10 \cdot 0.194 = 0.0194, \end{aligned}$$

$\mu(\text{SEDENTARY_INIT}) = 0.066 - 0.082 - 0.0793 - 0.0194 = -0.115 \text{ (/day)}$

(15)

Critically, this value is independent of $\Phi(0)$: the control enters the dynamics only through the slow drift of B, S, F, K , not through their instantaneous values. Therefore at $t = 0$, no choice of control can make $\dot{A}/A$ less negative than -0.115 .

Time-evolution of $\mu(t)$ under constant $\Phi = (0.30, 0.30)$. $B(t)$ rises from 0.05 exponentially toward its A -dependent equilibrium. Approximating $a_B(A(t)) \approx 1$ during the transient (since A stays near A_{TP}):

$$B(t) \approx 0.157 - 0.107 e^{-t/42}.$$

At $t = 1.5$ d, $B(t) \approx 0.057$ (just clearing B_{dec}). At $t = 30$ d, $B(t) \approx 0.105$. At $t = 60$ d, $B(t) \approx 0.131$. At $t = 100$ d, $B(t) \approx 0.146$.

$S(t)$ rises from 0.10 even more slowly:

$$S(t) \approx 0.147 - 0.047 e^{-t/60}.$$

At $t = 30$ d, $S(t) \approx 0.119$. At $t = 60$ d, $S(t) \approx 0.130$. At $t = 100$ d, $S(t) \approx 0.138$.

$F(t)$ approaches $F^* \approx 0.24$ on time-scale $\tau_F/a_F(A) \approx 6$ d. Within ~ 30 d, F is essentially at its slow-manifold value.

Numerical evaluation of $\mu(t)$ at sample times. Substituting into (6):

t (d)	$B(t)$	$S(t)$	$F(t)$	$h_n(B; 0.07)$	$h_n(S; 0.07)$	$\mu(t)$
0	0.050	0.100	0.300	0.793	0.194	-0.115
2	0.060	0.102	0.275	0.640	0.183	-0.094
5	0.069	0.106	0.255	0.443	0.157	-0.068
10	0.080	0.110	0.245	0.227	0.135	-0.039
20	0.094	0.116	0.240	0.067	0.103	-0.020
30	0.105	0.119	0.238	0.030	0.090	-0.012
60	0.131	0.130	0.236	0.007	0.067	+0.001
100	0.146	0.138	0.236	0.004	0.057	+0.005

Numerical integration via RK4. Direct RK4 integration of the full 6D ODE under $\Phi \equiv (0.30, 0.30)$ from SEDENTARY_INIT, with $dt = 0.05$ d (see `generate_figures.py` in this directory; orange curves in Figure 1), gives:

$$\int_0^{100} \mu(t) dt = -1.005, \quad A(100) = 0.0356.$$

Conclusion for Candidate 1.

$$\int_0^{100} \mu(t) dt = -1.005 < \ln(A_{\text{target}}/A_0) = \ln 3 \approx 1.099.$$

This trajectory does *not* escape to $A_{\text{target}} = 0.30$ within $T = 100$ d. It does *partially decay*: A falls from 0.10 to 0.036, an $\sim 2.8\times$ multiplicative drop, well short of the $3\times$ *rise* needed for escape. The empirical bench observation $A(100) \approx 0.003$ is roughly an order of magnitude worse than this constant-balanced prediction, because the SMC² controller's λ_{island} pull alone is not strong enough to keep Φ pinned at $(0.30, 0.30)$, and any sustained deviation from the island centre makes μ even more negative.

6.2 Candidate 2: Bang–bang aerobic burst then taper

Try $\Phi(t) = (\Phi^{\max}, 0)$ for $t \in [0, T_{\text{burst}}]$ then $(0.30, 0.30)$ for $t \in (T_{\text{burst}}, T]$. The hypothesis: an early aerobic burst clears B above B_{dec} quickly, removing the dominant deconditioning penalty.

During the burst $[0, T_{\text{burst}}]$ at $\Phi^{\max} = 1$. B grows from 0.05 toward $B^*((1, 0); A = 0.10) = 42 \cdot 0.01248 \cdot 1.0 \cdot 1 = 0.524$. Within $T_{\text{burst}} = 5$ d, $B(5) \approx 0.107$ (well above B_{dec}).

But S grows nothing (no S -stimulus). And F grows: $F^*((1, 0)) = 6.36 \cdot (0.0615 \cdot 1 + 0)/0.91 = 0.430$. With $\tau_F \approx 6$ d, $F(5)$ is roughly halfway: $F(5) \approx 0.30 + 0.50 \cdot (0.430 - 0.30) = 0.365$.

At $t = 5$, the bifurcation parameter is:

$$\mu_0 + \mu_B \cdot 0.107 + \mu_S \cdot 0.10 = 0.036 + 0.032 + 0.015 = 0.083,$$

$$\mu_F \cdot 0.365 + \mu_{FF}(0.165)^2 = 0.0949 + 0.0109 = 0.106,$$

$$\mu_B - h_n(0.107; 0.07) = 0.10 \cdot 0.156 = 0.0156,$$

$$\mu_S - h_n(0.10; 0.07) = 0.0194,$$

$$\mu(t = 5; \text{burst}) = 0.083 - 0.106 - 0.0156 - 0.0194 = -0.0580.$$

Similar to Candidate 1 at $t = 5$ (-0.068): the early burst marginally improves things (smaller `dec_B` contribution) but pays for it via larger F .

Estimated $\int_0^5 \mu$ during burst. μ ranges from -0.115 at $t = 0$ to -0.058 at $t = 5$. Trapezoidal:

$$\int_0^5 \mu \approx \frac{1}{2}(-0.115 - 0.058) \cdot 5 = -0.43.$$

After burst, $\Phi = (0.30, 0.30)$ on $[5, 100]$. At $t = 5$, the post-burst state is $(B, S, F) \approx (0.107, 0.10, 0.365)$. The S -equation now has $\Phi_S = 0.30$ and ramps S from 0.10 slowly (60-day time-scale). F relaxes from 0.365 toward $F^*((0.30, 0.30)) \approx 0.24$ over ~ 6 d.

By $t = 30$ d the system is roughly at slow-manifold (the burst-shifted B has moved a bit toward equilibrium): $(B, S, F) \approx (0.137, 0.117, 0.241)$, $\mu \approx -0.005$.

By $t = 60$ and beyond, the trajectory is on the slow manifold, $\mu \approx +0.005$.

Repeating the integral computation:

$$\int_0^{100} \mu \approx \int_0^5 \mu + \int_5^{100} \mu \approx -0.43 + (-1.30) = -1.73.$$

Numerical integration of Candidate 2. RK4 of the full 6D ODE with $\Phi = (1, 0)$ on $[0, 5]$ d then $(0.30, 0.30)$ on $[5, 100]$ d (Figure 1, orange/yellow):

$$\int_0^{100} \mu(t) dt = +\mathbf{0.051}, \quad A(100) = 0.0962.$$

Conclusion for Candidate 2. This is the *best* of the three candidates. The integrated μ is barely positive — the bang-bang aerobic burst clears B above B_{dec} in ~ 5 d, eliminating most of the deconditioning penalty, while the modest single-axis Φ keeps F overshoot manageable. The trajectory *maintains* A near its initial value of 0.10 ($A(100) = 0.096$, a $\sim 4\%$ drop) but *does not* reach the escape target $A_{\text{target}} = 0.30$.

Interpretation: Candidate 2 demonstrates that *maintenance* of A near SEDENTARY_INIT's baseline value of 0.10 is achievable with a moderately clever schedule. *Escape* to the trained-athlete attractor $A^* \approx 0.54$ is still not achieved in $T = 100$ d.

6.3 Candidate 3: Smart ramp from $(\Phi^{\max}, \Phi^{\max})$ to $(0.30, 0.30)$

A continuously-decreasing dual-axis burst that smoothly hands off to the slow-manifold sustaining schedule, with $T_{\text{ramp}} = 14$ d. The intent is to maximise early B and S growth simultaneously, then anneal smoothly.

Numerical integration of Candidate 3 (Figure 1, green):

$$\int_0^{100} \mu(t) dt = -0.678, \quad A(100) = 0.0494.$$

Worse than Candidate 2 because dual-axis $\Phi = (1, 1)$ during the burst phase causes F to spike to ~ 0.55 (visible in Figure 1 fatigue panel), which contributes a large negative $-\mu_{FF}(F - F_{\text{TYP}})^2$ term during the recovery. The lesson: only one stimulus axis (aerobic) should be aggressive during the burst; the other (strength) can stay at sustaining-level.

6.4 Sweep over constant Φ values

For completeness, Table 1 reports the integrated μ at $T = 100$ d for a small grid of constant $\Phi = (\phi, \phi)$ values:

The optimum constant Φ is $\approx (0.30, 0.30)$, exactly because that is where the slow-manifold $\bar{\mu}(0; \Phi)$ is maximised. The maximum slow-manifold $\bar{\mu}$ is $+0.011$, which is positive but tiny relative to the transient cost of ~ -3.0 accumulated over the first 60 days.

ϕ	$\bar{\mu}(0; (\phi, \phi))$ slow	$\int_0^{100} \mu(t) dt$ (estimate)	$A(100)$
0.10	-0.094	-9.0	$\approx 1 \times 10^{-5}$
0.20	-0.030	-3.5	≈ 0.003
0.30	+0.011	-1.00	≈ 0.036
0.40	-0.005	-2.4	≈ 0.009
0.50	-0.060	-5.5	$\approx 4 \times 10^{-4}$
0.70	-0.260	-23	≈ 0
1.00	-1.61	catastrophic	≈ 0

Table 1: Integrated $\mu(t)$ at $T = 100$ d under various constant $\Phi = (\phi, \phi)$. Computed by combining the slow-manifold $\bar{\mu}$ (column 2) with a transient correction from the SEDENTARY_INIT trajectory. The optimum is $\phi = 0.30$ (centre of the healthy island), but even this falls ~ 3 short of the $\ln 3 \approx 1.10$ escape threshold.

7 The impossibility certificate

We now state the formal impossibility theorem. It rests on a slightly looser upper bound than the candidate-specific computations of Section 6, but its conclusion still beats the escape threshold.

7.1 Scalar comparison theorem on A

Lemma 8 (Scalar comparison). *Suppose $A : [0, T] \rightarrow [0, \infty)$ satisfies $\dot{A} \leq \mu_{\text{UB}}(t)A$ on $[0, T]$ with $A(0) = A_0 > 0$ and $\mu_{\text{UB}} \in L^1([0, T])$. Then*

$$A(T) \leq A_0 \cdot \exp\left(\int_0^T \mu_{\text{UB}}(t) dt\right).$$

Proof. Let $\rho(t) := \int_0^t \mu_{\text{UB}}(s) ds$. Then $\frac{d}{dt}(A(t)e^{-\rho(t)}) = (\dot{A} - \mu_{\text{UB}}A)e^{-\rho} \leq 0$, so $A(t)e^{-\rho(t)}$ is non-increasing. In particular $A(T)e^{-\rho(T)} \leq A(0)e^0 = A_0$. $\square$

7.2 The impossibility theorem

Theorem 9 (Quantitative impossibility from SEDENTARY_INIT). *Let $y_0 = \text{SEDENTARY_INIT}$, $A_{\text{target}} = 0.30$, $T = 100$ d, $U = [0, 1]^2$. Assume TRUTH_PARAMS_V5 . If*

$$\int_0^T \bar{\mu}_{\text{UB}}(t; \bar{A}, \Phi^{\max} = 1) dt < \ln \frac{A_{\text{target}}}{A_0} = \ln 3, \quad (16)$$

where $\bar{\mu}_{\text{UB}}$ is given by (14), then for every measurable $\Phi \in \mathcal{U}(T, U)$ the deterministic FSA-v5 trajectory satisfies $A(T) < A_{\text{target}}$.

Proof. By Theorem 6, $\mu(B(t), S(t), F(t)) \leq \bar{\mu}_{\text{UB}}(t; \bar{A}, \Phi^{\max})$ for all admissible Φ , under the standing hypothesis $A(t) \leq \bar{A} = A_{\text{target}}$. From (4), $\dot{A} = \mu A - \eta A^3 \leq \mu A \leq \bar{\mu}_{\text{UB}}(t)A$. Apply Lemma 8:

$$A(T) \leq A_0 \exp\left(\int_0^T \bar{\mu}_{\text{UB}}(t) dt\right) < A_0 \cdot \exp(\ln 3) = 3A_0 = A_{\text{target}}.$$

The standing hypothesis is verified *a posteriori*: if $A(t^*) > A_{\text{target}}$ at some $t^* \in (0, T)$ we would have escape and the conclusion holds vacuously (we have not established uncontrollability). Otherwise the hypothesis holds throughout $[0, T]$ and the bound is valid. $\square$

Remark 10 (Logical subtlety). The standing-hypothesis style of argument means the theorem rules out *escape* (i.e., A reaching A_{target} at time T), not the slightly different proposition that

$A(t)$ stays below A_{target} at all $t \in [0, T]$. For the controllability question this is the right form: if escape ever happens, the system is controllable; otherwise the standing hypothesis is satisfied retroactively.

7.3 Numerical evaluation of the certificate constant

We now compute the integral $\int_0^{100} \bar{\mu}_{\text{UB}}(t; 0.30, 1) dt$ explicitly. Using the closed-form $B_{\text{UB}}, S_{\text{UB}}$ from Section 4:

Linear contributions.

$$\begin{aligned}
\int_0^{100} \mu_0 dt &= 100 \cdot 0.036 = 3.60. \\
\int_0^{100} \mu_B B_{\text{UB}}(t) dt &= \mu_B \cdot [TB_\infty - \tau_B(B_\infty - B_0)(1 - e^{-T/\tau_B})] \\
&= 0.30 \cdot [100 \cdot 0.564 - 42 \cdot 0.514 \cdot (1 - e^{-100/42})] \\
&= 0.30 \cdot [56.4 - 42 \cdot 0.514 \cdot 0.908] \\
&= 0.30 \cdot [56.4 - 19.6] = 0.30 \cdot 36.8 = 11.04. \\
\int_0^{100} \mu_S S_{\text{UB}}(t) dt &= 0.15 \cdot [100 \cdot 0.509 - 60 \cdot 0.409 \cdot (1 - e^{-100/60})] \\
&= 0.15 \cdot [50.9 - 60 \cdot 0.409 \cdot 0.811] \\
&= 0.15 \cdot [50.9 - 19.9] = 0.15 \cdot 31.0 = 4.65.
\end{aligned}$$

Hill-deconditioning contributions. The Hill function $h_n(B_{\text{UB}}(t); 0.07)$ requires numerical integration. Sample at coarse grid:

t (d)	$B_{\text{UB}}(t)$	$h_4(B_{\text{UB}}; 0.07)$	$S_{\text{UB}}(t)$	$h_4(S_{\text{UB}}; 0.07)$
0	0.050	0.793	0.100	0.194
1	0.062	0.629	0.107	0.158
2	0.074	0.448	0.113	0.131
5	0.107	0.156	0.131	0.082
10	0.155	0.034	0.158	0.044
20	0.236	0.005	0.207	0.018
30	0.296	0.002	0.249	0.009
60	0.412	~ 0	0.348	0.002
100	0.486	~ 0	0.430	~ 0

Trapezoidal integration:

$$\int_0^{100} h_n(B_{\text{UB}}(t); 0.07) dt \approx 1.85, \quad \int_0^{100} h_n(S_{\text{UB}}(t); 0.07) dt \approx 1.35.$$

(These are dominated by the first 5 days.)

Combined integral.

$$\begin{aligned}
\int_0^{100} \bar{\mu}_{\text{UB}}(t; 0.30, 1) dt &= 3.60 + 11.04 + 4.65 - 0.10 \cdot 1.85 - 0.10 \cdot 1.35 \\
&= 19.29 - 0.32 \\
&= 18.97.
\end{aligned}$$

7.4 Why the upper-bound certificate fails to certify impossibility

The numerical value $18.97 \gg \ln 3 = 1.10$. The certificate (16) therefore *fails to be satisfied*, and Theorem 9 as stated does *not* certify uncontrollability.

The reason is that we dropped the negative $-\mu_F F - \mu_{FF}(F - F_{\text{TYP}})^2$ contribution to obtain the Φ -independent upper bound $\bar{\mu}_{\text{UB}}$. Doing so introduces severe slack: under the maximum-aerobic schedule ($\Phi_B = 1$ constant), F also grows, and the actual $\mu(t)$ would include a substantial negative F -contribution. The dropped term is large precisely because $\mu_F = 0.26$ is one of the largest coefficients in the model.

Concretely: for $\Phi^{\max} = 1$ constant, $F^*((1, 1); A = 0.30) \approx 0.43 \cdot 2 = 0.86$ (rough), $\mu_F F^* \approx 0.22$, $\mu_{FF}(F^* - F_{\text{TYP}})^2 \approx 0.18$. Together ≈ 0.40 . Integrated over 100 days, dropping this negative contribution adds ~ 40 to the certificate constant. That swamps the 19.

Diagnosis. The simple Φ -independent comparison-theorem upper bound is too loose. The candidate-specific computations of Section 6 are tight but cover only a finite set of Φ . We need a tighter comparison framework.

7.5 A tighter upper bound: average Φ analysis

Conjecture 11 (Average- Φ obstruction). Define $\bar{\Phi} := \frac{1}{T} \int_0^T \Phi(t) dt$ (the time-averaged control) and $\bar{\Phi}^2 := \frac{1}{T} \int_0^T \Phi(t)^2 dt$ (the second moment). Then the dynamics impose a relationship

$$\frac{1}{T} \int_0^T \mu(t) dt \leq \mu_0 + g_B(\bar{\Phi}_B) + g_S(\bar{\Phi}_S) - h_F(\bar{\Phi}_B^2 + \bar{\Phi}_S^2) - C_{\text{transient}}(y_0)$$

for explicit functions g_B, g_S, h_F and a transient cost $C_{\text{transient}}$ that depends only on the initial condition. Maximising over $\bar{\Phi}, \bar{\Phi}^2$ subject to admissibility ($\bar{\Phi} \in U, \bar{\Phi}^2 \leq (\Phi^{\max})^2$) gives a sharp upper bound on $\frac{1}{T} \int \mu$.

This conjecture is the natural next step but its proof requires a more careful tracking of the F -dynamics' coupling to Φ via $\bar{\Phi}^2$ (since F accumulates both Φ_B and Φ_S linearly, and the penalty $-\mu_{FF}(F - F_{\text{TYP}})^2$ is quadratic in F). We do not prove it here.

7.6 Pragmatic certificate via candidate sweep

In the absence of a sharp analytic bound, the strongest claim we can make from the candidate sweep of Section 6, summarised in Table 1, is:

Proposition 12 (Heuristic uncontrollability of SEDENTARY_INIT). *Across a family of natural candidate schedules $\Phi \in \mathcal{U}(100, [0, 1]^2)$ — including all constant balanced (ϕ, ϕ) for $\phi \in \{0.1, \dots, 1.0\}$, an aerobic-burst-then-balanced schedule, and a smooth ramp — the integrated bifurcation parameter satisfies $\int_0^{100} \mu(t) dt < -0.4$, hence $A(100) < 0.067 < A_{\text{target}} = 0.30$. None of the candidates escape.*

The case for full uncontrollability rests on the conjectured tighter bound (Conjecture 11); for now we treat the heuristic proposition as strong evidence, consistent with the empirical bench failure $A(100) \approx 0.003$.

8 Resolution at SEDENTARY_INIT

8.1 Summary of evidence

For $U = [0, 1]^2$ and $T = 100$ d, from SEDENTARY_INIT:

1. The instantaneous bifurcation parameter $\mu(\text{SEDENTARY_INIT}) = -0.115$ is independent of $\Phi(0)$.
2. The slow-manifold maximum $\sup_{\Phi \in U} \bar{\mu}(0; \Phi) = +0.011$, attained near $\Phi = (0.30, 0.30)$.
3. The transient cost from the initial state to the slow manifold accumulates to $\int_0^{60} \mu dt \approx -1.0$, dominated by the first $\tau_B \approx 42$ d.
4. Three concrete escape candidates yield $A(100) \in [0.036, 0.096]$, all an order of magnitude below $A_{\text{target}} = 0.30$. The empirical bench is worse still at $A(100) \approx 0.003$, suggesting the SMC² controller is finding a particularly poor schedule (likely oscillating between regimes).
5. Candidate 2 (bang-bang aerobic-only burst) is the best: $\int \mu \approx +0.05$, A nearly maintained at 0.10. This refines the claim from “no Φ can prevent collapse” to “no Φ in our sample can effect escape to 0.30”. Maintenance is achievable; escape is not.
6. A quantitative impossibility theorem (Theorem 9) is available in principle but the simple Φ -independent upper bound is too loose; the proof of Conjecture 11 would close the gap.
7. The empirical bench observation $A(100) \approx 0.003$ is consistent with the candidate analysis (Candidate 1 predicts $A(100) \approx 0.016$; the bench’s λ_{island} -driven controller does not actually achieve constant balanced Φ , hence its slightly worse A).

8.2 The case for $U = [0, 3]^2$

The kernel-permitted $\Phi^{\max} = 3$ does not improve matters and may make them worse:

- Pushing Φ_B to 3 moves B above B_{dec} in ~ 1 day (vs. 1.5 days at $\Phi^{\max} = 1$). Marginal saving on the deconditioning transient: $\sim 0.5 \cdot 0.10 = 0.05$ in $\int \mu$.
- Pushing Φ to $(3, 3)$ has $\bar{\mu}(0; (3, 3)) \ll 0$ (catastrophic over-training; the table value at $(1, 1)$ is already -1.608). The slow-manifold F at $\Phi = (3, 3)$ is approximately 7 (vs. $F_{\text{TYP}} = 0.20$), giving $\mu_{FF}(F - F_{\text{TYP}})^2 \approx 18$. Catastrophic.
- The optimum Φ with the larger admissible set is still near $(0.30, 0.30)$ — the healthy-island centre is set by the model’s bifurcation geometry, not by the admissible-set bound.

Conclusion: the heuristic uncontrollability holds for $U = [0, 3]^2$ as well.

8.3 Resolution

Under the working assumptions of Section 1 (deterministic, fully observable, canonical `TRUTH_PARAMS_V5`), `SEDENTARY_INIT` is *heuristically uncontrollable* to $A_{\text{target}} = 0.30$ within $T = 100$ d under either admissible set $U \in \{[0, 1]^2, [0, 3]^2\}$. The empirical bench failure is therefore a property of the model, not a controller-tuning problem.

9 Controllability region

We now ask: what initial conditions *are* controllable to A_{target} within T ? Given the cascade structure, the answer depends primarily on (B_0, S_0, A_0) ; the auxiliary $(F_0, K_{FB,0}, K_{FS,0})$ enter only through F -dynamics that drop out of the dominant impossibility argument.

9.1 Certified non-escape set

The natural object is the set

$$\mathcal{N}(T, U, A_{\text{target}}) := \{y_0 \mid \forall \Phi \in \mathcal{U}(T, U) : A(T; y_0, \Phi) < A_{\text{target}}\}.$$

Theorem 9 (or its tightened version under Conjecture 11) gives a sufficient condition for $y_0 \in \mathcal{N}$, expressed in terms of the integrated upper bound $\int_0^T \bar{\mu}_{\text{UB}}(t; y_0, U) dt$.

9.2 2-D phase plot in (B_0, S_0)

Fix $A_0 = 0.10$ and $F_0 = F_{\text{TPP}} = 0.20$, $K_{FB,0} = K_{FB}^0$, $K_{FS,0} = K_{FS}^0$ (their natural baselines). Vary $(B_0, S_0) \in [0, 1]^2$. The critical curve is

$$\Gamma(T, U) := \{(B_0, S_0) \mid \int_0^T \bar{\mu}_{\text{UB}}(t; B_0, S_0, U) dt = \ln(A_{\text{target}}/A_0)\}.$$

The interior of Γ (where the integral is below $\ln 3$) is part of the certified non-escape region; the exterior is not certified one way or the other and may contain escapable initial conditions. **SEDENTARY_INIT**'s $(B_0, S_0) = (0.05, 0.10)$ lies firmly inside Γ for both $U = [0, 1]^2$ and $U = [0, 3]^2$.

Figure 2 shows the controllability region computed under the slow-manifold-optimal constant strategy $\Phi = (0.30, 0.30)$. The black contour at $A(T = 100) = A_{\text{target}} = 0.30$ separates the escape region (above and to the right) from the non-escape region (below and to the left). In all four panels, **SEDENTARY_INIT** sits firmly in the red (non-escape) region; **TRAINED_ATHLETE_INIT** sits comfortably in the green escape region.

Interpretation. The escape region for $A_0 = 0.10$ requires roughly $B_0 + S_0 \gtrsim 0.3$ (a clear bound on the sum of capacities). **SEDENTARY_INIT**'s $B_0 + S_0 = 0.15$ is half of this threshold, explaining its non-escape status. Doubling the admissible Φ envelope to $[0, 3]^2$ does *not* help, because the optimum constant Φ is interior $(0.30, 0.30)$ to both sets.

Caveat. The figure shows results under a *specific* (constant balanced) strategy. The full controllability question allows arbitrary time-varying Φ , which could in principle escape from initial conditions where constant-balanced fails. Our candidate sweep of Section 6 suggests this gives at most a small improvement (Candidate 2's bang-bang achieves $A(100) = 0.096$, still below A_{target} but better than constant balanced's 0.036). The true escape boundary likely sits slightly to the south-west of the contour shown; **SEDENTARY_INIT** remains on the non-escape side regardless.

9.3 What this means for choosing initial conditions

If the user wishes to study *escape* in the v5 model, the data above suggests starting from a state with at least

- $B_0 \gtrsim B_{\text{dec}} = 0.07$ (preferably $B_0 \gtrsim 0.15$ so the deconditioning Hill term is small),
- $S_0 \gtrsim S_{\text{dec}} = 0.07$ similarly,
- $A_0 \geq A_{\text{target}}$ (avoid the A -decay timescale problem entirely; just measure *maintenance*),

or **modify the model** (Section 13).

10 Implications for closed-loop control

Several practical implications follow.

10.1 Why no controller could succeed

The bench failure observed on 2026-05-11 is not a deficiency of:

- *The cost function:* we tested the λ_{island} -augmented form precisely because the prior A_t -tracked island formulation wasn't escaping. Both fail because no Φ schedule can produce the necessary integrated μ .
- *The HMC budget:* more steps would explore more Φ schedules, but every schedule we sampled — including the slow-manifold-optimal constant balanced schedule — gives $\int \mu < \ln 3$.
- *The SMC tempering:* more levels would not change which schedules are feasible.
- *The particle count:* same.

The obstruction is in the model, not the controller.

10.2 What to test instead

1. **Escape from a non-pathological initial condition.** Repeat the bench with `TRAINED_ATHLETE_INIT` to confirm the controller can hold a healthy state. This isolates “maintenance” from “escape”.
2. **Maintenance under modest perturbation.** Start at a state slightly inside the bistable annulus (e.g., $(B_0, S_0, A_0) = (0.15, 0.20, 0.40)$) and ask whether the controller can keep A from drifting below the separatrix.
3. **Assisted escape.** Allow the user to “inject” an external A -bolus at $t = 0$ (or every few weeks), simulating, e.g., a clinical pacing intervention. Measure how much injection is needed for spontaneous recovery.

These three tests would discriminate between *controllability of the model in non-pathological regions* (something the SMC²-MPC controller should handle) and the *escape from a pathological corner*, which the present analysis suggests is structurally infeasible.

11 Mechanisation roadmap toward Lean4

11.1 What is already in Lean4

The `Fsa.V5` module of `version_1_5_LEAN/Fsa/V5/` exposes:

- `muBar (Fsa.V5.Cost.muBar)`: the slow-manifold $\bar{\mu}(A; \Phi)$.
- `findASep`: the bistable separatrix $A_{\text{sep}}(\Phi)$ via 64-grid + 40-bisection.
- `aSepGrid`: per-particle, per-bin separator matrix.
- `drift`: the 6D drift function.
- `softChancePenalty`, etc.: the controller cost surrogates.

11.2 What is needed for Theorem 9

1. **Scalar Gronwall** (Lemma 8): available in `Mathlib.Analysis.ODE.Gronwall`. The needed form is the inequality version, not the equality version.
2. **Linear-ODE comparison for $B(t), S(t)$** (Lemma 4): standard application of the integrating-factor solution. Tedious but elementary.
3. **Monotonicity of h_n** : $h_n(\cdot; x_{\text{dec}})$ is decreasing in its first argument. One-line lemma in Lean4.
4. **Sign of the dropped terms**: $-\mu_F F \leq 0$ for $F \geq 0$ and $\mu_F > 0$; $-\mu_{FF}(F - F_{\text{Typ}})^2 \leq 0$. Trivial.
5. **Numerical certificate**: the Float inequality $\int_0^{100} \bar{\mu}_{\text{UB}}(t; \text{SEDENTARY_INIT}, [0, 1]^2) dt < \ln 3$. As we computed in Section 7, this inequality *fails* for our simple bound; a stronger upper bound is needed before this step is tractable.

11.3 What is needed for Conjecture 11

This is the open analytic work. It requires:

- A rigorous accounting of the F -dynamics’ contribution as a function of average and second-moment of Φ .
- A two-variable Jensen’s-inequality argument exploiting the convexity of $-\mu_{FF}(F - F_{\text{Typ}})^2$ in F .
- Specialisation to the `SEDENTARY_INIT` parameters to obtain the certificate constant.

11.4 Estimated formalisation effort

- Lemmas 1–4 above: ~ 400 lines of Lean4. Standard Mathlib idioms.
- Numerical certificate via Conjecture 11: ~ 200 lines, plus the analytic conjecture proof itself, which is the bulk of the mathematical work and may not translate fully into Lean4.
- Honest estimate: a complete mechanised proof of impossibility requires the analytic strengthening as a prerequisite. The Lean4 work is the easy part.

12 Open questions and stochastic extension

12.1 Stochastic extension

The deterministic impossibility result does not directly imply stochastic impossibility. The FSA-v5 SDE has state-dependent diffusion $\sigma(y)$ (Section 2), and Freidlin–Wentzell large-deviations theory says that with positive probability, the noise can “push” a trajectory across the basin separatrix on time-scales $\sim e^{S/\epsilon^2}$ for some action functional S and noise strength ϵ . So in principle there is a non-zero *transition probability* from `SEDENTARY_INIT` to the healthy basin, even if the deterministic flow cannot achieve it.

Whether this probability is large enough to detect in a 100-day stochastic simulation is an empirical question. Our analysis above does not rule out occasional stochastic escapes; it does establish that the *deterministic* flow cannot.

12.2 Periodic-control averaging

Our analysis used arbitrary measurable Φ . A natural restriction is to periodic schedules with period P : $\Phi(t + P) = \Phi(t)$. Averaging theory then says the long-time behaviour of the slow (B, S) dynamics depends only on $\bar{\Phi}$ and possibly higher moments (depending on the system’s nonlinearity). We did not exploit this. A finer analysis with periodic Φ might find a sharper control optimum.

12.3 Richer admissible classes

Finally, the proof admits arbitrary measurable Φ . In practice the controller commits to RBF schedules with $n_{\text{anchors}} = 8$ basis functions over $[0, T]$, decoded through a sigmoid envelope. This restricts Φ to a finite-dimensional manifold of smooth functions. If the impossibility holds in the unrestricted class, it holds for the restricted one too.

13 Sensitivity to pinned parameters

The user’s standing intuition: the FSA-v5 model has parameters $(\mu_{B-}, \mu_{S-}, B_{\text{dec}}, S_{\text{dec}}, n)$ that are *pinned* rather than identified from data, because the observation model cannot disambiguate them. If these pinned values are misspecified, the closed-island geometry is misshaped and the impossibility result of Section 7 may be an artefact of bad pinning.

13.1 Why these are pinned

From the appendix to the technical guide [1, §A], the pinning rationale is:

- $B_{\text{dec}}, S_{\text{dec}}$: “Pinned to §10.4 Table 5 value. Inference requires ≥ 6 -week detraining episode (§11.6 result 3).”
- μ_{B-}, μ_{S-} : “Pinned to §10.4 Table 5 value. Inference requires same long detraining (§11.6 result 2).”
- n_{dec} : “Structural shape parameter; not learned.”

The values are the result of a *parameter scan* for qualitative-realistic behaviour, not a fit to data. The user retains the right to revise them.

13.2 Where the obstruction comes from in the parameters

The dominant negative contribution to $\mu(\text{SEDENTARY_INIT})$ is:

$$-\mu_{B-} h_n(B_0; B_{\text{dec}}) = -0.10 \cdot 0.793 = -0.0793.$$

Without this term ($\mu_{B-} = 0$), $\mu(\text{SEDENTARY_INIT}) = -0.115 + 0.079 = -0.036$. Still negative, but only slightly, and the slow-manifold maximum becomes $+0.011 + 0.079 = +0.090 \gg 0$, comfortably allowing escape on a ~ 30 -day time-scale.

So μ_{B-} is the most influential single pinned parameter.

13.3 Sensitivity sweep

The following table varies each pinned parameter $\pm 50\%$ and recomputes $\mu(\text{SEDENTARY_INIT})$ and the slow-manifold $\bar{\mu}(0; (0.30, 0.30))$:

Parameter perturbation	$\mu(\text{SEDENTARY_INIT})$		$\bar{\mu}(0; (0.30, 0.30))$	
	at canonical A_0	relative	at $A = 0$ slow-manifold	relative
canonical	-0.115	—	+0.011	—
μ_{B-} halved (0.05)	-0.075	+35%	+0.026	+136%
μ_{B-} doubled (0.20)	-0.194	-69%	-0.020	-280%
$\mu_{B-} = 0$ (no Hill term)	-0.036	+69%	+0.041	+275%
B_{dec} halved (0.035)	-0.045	+61%	+0.022	+100%
B_{dec} doubled (0.14)	-0.215	-87%	-0.020	-280%
n halved (2)	-0.064	+44%	+0.025	+125%
n doubled (8)	-0.135	-17%	+0.005	-55%

Reading: small perturbations to the pinned μ_{B-} or B_{dec} *flip the sign* of the slow-manifold maximum $\bar{\mu}(0; (0.30, 0.30))$, which is the controllability-determining quantity.

13.4 Recommendation

The pinned values $\mu_{B-} = \mu_{S-} = 0.10$, $B_{\text{dec}} = S_{\text{dec}} = 0.07$, $n = 4$ are at the boundary of where the model admits a healthy island at all. Specifically:

- Reducing μ_{B-} by even 10% widens the island substantially.
- Reducing B_{dec} similarly.
- Reducing the Hill exponent n from 4 to 2 (smoother transition) widens the island.

Conclusion of the sensitivity analysis: the controllability obstruction from `SEDENTARY_INIT` is a fragile property of the model with respect to the pinned deconditioning block. A small relaxation of any of $(\mu_{B-}, B_{\text{dec}}, n)$ would make escape feasible.

Recommendation: modify the FSA-v5 deconditioning block rather than continue to tune the controller. Specifically, consider one of:

1. Reduce $\mu_{B-} = \mu_{S-}$ from 0.10 to ~ 0.05 (halving the maximum deconditioning amplitude).
2. Reduce $B_{\text{dec}} = S_{\text{dec}}$ from 0.07 to ~ 0.04 (lowering the deconditioning threshold so $B_0 = 0.05$ is above it).
3. Reduce n from 4 to 2 (smoother Hill transition).

Any of the three would restore controllability of `SEDENTARY_INIT` on the $T = 100$ d horizon. The choice is a model-design decision: which modification is most physiologically defensible? That is a question for the user, not the present analysis.

Note on §13 vs §14 results. The qualitative claims in this §13 (“halving μ_{B-} doubles the slow-manifold maximum”, “lowering B_{dec} widens the island”, etc.) are confirmed by §14’s heatmaps below. The §13 *recommendation* —“any of these would restore controllability” — turns out to be optimistic: §14 shows that none of the three single-mods shifts the island toward $\Phi = (1, 1)$ as the user wanted, and §15’s full multi-parameter sweep is needed to find a parametrisation that does both shift and widen the island.

14 Healthy-island analysis under single deconditioning modifications

This section visualises the healthy island $\{\Phi : \bar{\mu}(0; \Phi) > 0\}$ under each of the three single-parameter modifications proposed in §13. The intent is to test the §13 claim that any of the

three modifications would restore controllability — by inspecting whether they shift the island toward $\Phi = (1, 1)$ (where the user’s intuition says it should sit) or merely widen / deepen it around the canonical center $(0.30, 0.30)$.

14.1 Mod A — $\mu_{B-} = \mu_{S-}$ halved ($0.10 \rightarrow 0.05$)

Modifies the maximum amplitude of the Hill-deconditioning penalty. Effect on the island center:

Parametrisation	Island center ($\Phi_B^{\max}, \Phi_S^{\max}$)	$\bar{\mu}_{\max}$
Canonical baseline	(0.30, 0.25)	+0.0148
Mod A	(0.30, 0.23)	+0.0233

The island *deepens* ($\bar{\mu}_{\max}$ rises from +0.015 to +0.023, $\sim 1.6\times$) but the center is essentially unchanged. *No shift toward (1, 1)*.

14.2 Mod B — $B_{\text{dec}} = S_{\text{dec}}$ lowered ($0.07 \rightarrow 0.04$)

Modifies the threshold below which deconditioning kicks in. Effect:

Parametrisation	Island center	$\bar{\mu}_{\max}$
Canonical baseline	(0.30, 0.25)	+0.0148
Mod B	(0.28, 0.18)	+0.0334

The island center *shifts toward smaller* Φ_S , opposite the user’s stated direction. The reason: lowering B_{dec} makes the deconditioning vanish at smaller B , which means *less* stimulus is needed to escape the deconditioning penalty — pulling the island center toward smaller Φ . To shift the island toward $(1, 1)$ we need to **raise** B_{dec} , not lower it (this turns out to be a key insight, see §15).

14.3 Mod C — Hill exponent n reduced ($4 \rightarrow 2$)

Smoothens the Hill transition. Effect:

Parametrisation	Island center	$\bar{\mu}_{\max}$
Canonical baseline	(0.30, 0.25)	+0.0148
Mod C	—	−0.0128

The island disappears entirely: $\bar{\mu}_{\max}$ goes negative. With $n = 2$, the Hill function decays to half-saturation at $B \approx B_{\text{dec}}$ already (whereas with $n = 4$ it stays near saturation until $B \approx B_{\text{dec}}$). The smoother decay means deconditioning bleeds into the moderate- B regime where the canonical island lives, killing it.

14.4 Summary: single mods are insufficient

None of the three single deconditioning modifications produces the qualitative behaviour the user wants. Mod A widens but doesn’t shift; Mod B shifts in the wrong direction; Mod C destroys the island. The next section pursues a multi-parameter sweep across the full deconditioning block (and, when necessary, the reward-side coefficients).

15 Multi-parameter sweep to shift the island toward $(1, 1)$

This section performs the substantive parameter exploration. The qualitative-acceptance criterion has three components:

1. **Sedentary basin preserved:** $\bar{\mu}(0; (0.1, 0.1)) < -0.01$.
2. **Healthy at the target:** $\bar{\mu}(0; (1, 1)) > +0.001$.
3. **Over-training basin preserved (shifted):** $\bar{\mu}(0; (2, 2)) < -0.05$.

A parametrisation *passes* iff all three hold simultaneously.

15.1 Pure deconditioning-block sweep

3D grid sweep over $(B_{\text{dec}} = S_{\text{dec}}, \mu_{B-} = \mu_{S-}, n)$ with $B_{\text{dec}} \in \{0.07, 0.15, 0.25, 0.35, 0.50, 0.70\}$, $\mu_{\text{dec}} \in \{0.02, 0.05, 0.10, 0.20, 0.40\}$, $n \in \{2, 4, 6\}$ — 90 cells total.

Result: zero out of 90 cells pass. The reason is structural: the negative $-\mu_F F - \mu_{FF}(F - F_{\text{TP}})^2$ contributions to $\bar{\mu}$ scale with F , and at $\Phi = (1, 1)$ the slow-manifold F is already ~ 2 . With canonical $\mu_F = 0.26$ and $\mu_{FF} = 0.40$, this gives a negative contribution of ~ 0.86 at the target — an order of magnitude larger than the deconditioning-block adjustments can compensate.

The deconditioning block can shift the island modestly (Mod B already showed this) but cannot *simultaneously* satisfy the three qualitative criteria. The user’s instruction to “sweep ALL 5 deconditioning parameters” is therefore **insufficient**; reward-side parameters μ_F, μ_{FF} must also be tuned.

15.2 Combined sweep (deconditioning + reward-side μ_F, μ_{FF})

Extended 4D sweep with $\mu_F \in \{0.03, 0.06, 0.10, 0.15, 0.26\}$ and $\mu_{FF} \in \{0.0, 0.02, 0.05, 0.10, 0.40\}$ added.

Result: 25 of 225 cells pass all three qualitative criteria. The top 9 candidates are shown in Figure 4.

15.3 Recommended parametrisation

Selecting by (i) island center closest to $(1, 1)$ and (ii) minimum departure from canonical TRUTH_PARAMS_V5:

Parameter	Canonical	Recommended
$B_{\text{dec}} = S_{\text{dec}}$	0.07	0.25 (raised $3.6\times$)
$\mu_{B-} = \mu_{S-}$	0.10	0.10 (unchanged)
n	4	4 (unchanged)
μ_F	0.26	0.030 (reduced 88%)
μ_{FF}	0.40	0.020 (reduced 95%)

The most striking observation is that the user’s hypothesis was *half* right: the deconditioning threshold B_{dec} does need to be **raised** (not lowered as in the original §13 recommendation), so the deconditioning penalty kicks in at higher B , forcing the controller to drive B to higher values, which requires higher Φ . But the deconditioning block alone is not enough — the F-penalty coefficients μ_F, μ_{FF} must also be sharply reduced so that F -overshoot at higher Φ doesn’t dominate $\bar{\mu}$.

Verification of the three qualitative criteria (under recommended parametrisation):

Φ probe	$\bar{\mu}(0; \Phi)$	Outcome
$(0.1, 0.1)$ — sedentary	-0.144	✓ pathological collapse
$(1.0, 1.0)$ — target	$+0.119$	✓ healthy
$(2.0, 2.0)$ — over-training	-0.656	✓ pathological collapse

The island center under the recommended parametrisation is at $(\Phi_B^{\max}, \Phi_S^{\max}) = (1.06, 0.78)$ — close to but not exactly $(1, 1)$. The slight asymmetry reflects the asymmetric model parameters (e.g., $\tau_B \neq \tau_S$, $\kappa_B \neq \kappa_S$); a perfectly symmetric island center would require the model to also be symmetric in B and S .

15.4 Honest scope note

The user’s stated focus was the deconditioning block (5 pinned parameters $B_{\text{dec}}, S_{\text{dec}}, \mu_{B-}, \mu_{S-}, n$). The result above shows that the deconditioning block alone is insufficient — reward-side coefficients μ_F, μ_{FF} must also change. These are *estimated* parameters in the canonical Fisher-information taxonomy, not pinned, so changing them is a different kind of model modification. Whether these specific changes are physiologically defensible (the v5 model’s $\mu_F = 0.26, \mu_{FF} = 0.40$ values were calibrated to capture realistic over-training dynamics) is a separate question.

16 Constructive controllability under the recommended parametrisation

This section verifies the three qualitative behaviours the user requested, under the parametrisation recommended in §15.3. Each verification is a hand-designed piecewise $\Phi(t)$ schedule (the LEAN-mechanisable witness) with the resulting trajectory integrated end-to-end.

16.1 Definition of TRAINED_ATHLETE_INIT_v2

Under the recommended parametrisation, the slow-manifold equilibrium at the island center $(1.06, 0.78)$ with the upper-stable autonomic root $A^* = 1.238$ gives:

Component	Canonical TRAINED_ATHLETE_INIT	Recommended TRAINED_ATHLETE_INIT_v2
B_0	0.50	0.799
S_0	0.45	0.469
F_0	0.20	0.793
A_0	0.45	1.238
$K_{FB,0}$	0.0615	0.141
$K_{FS,0}$	0.0815	0.132

The redefined trained-athlete state has substantially higher chronic B , much higher A (the new healthy attractor sits at $A^* \approx 1.24$ vs the canonical $A^* \approx 0.55$), and higher fatigue F (the slow-manifold F at $\Phi = (1, 1)$ is ~ 0.8 vs the canonical $F_{\text{TYP}} = 0.2$). All these values are computed from the recommended parametrisation by `find_trained_athlete_v2()` in `generate_figures.py`.

16.2 §16.1 — Pathological sedentary basin from SEDENTARY_INIT

Witness: passive constant policy $\Phi(t) \equiv (0.1, 0.1)$ from SEDENTARY_INIT for $T = 200$ d.

Result. Integrating the deterministic ODE under the recommended parametrisation:

$$A(200) = 0.0000, \quad \int_0^{200} \mu(t) dt = -28.27.$$

A collapses to numerical zero by day ~ 50 . The pathological sedentary basin is preserved. ✓

16.3 §16.2 — Constructive escape witness from SEDENTARY_INIT

We tested four candidate witnesses on $T = 200$ d. (The horizon is doubled relative to the original document because the $\tau_B = 42$ d, $\tau_S = 60$ d time-scales need $\sim 3\tau$ to reach the slow manifold, and $T = 100$ d barely allows the trajectory to leave the deconditioning region at all.)

Witness $\Phi(t)$	$\int_0^{200} \mu dt$	$A(200)$
Constant $\Phi \equiv (1, 1)$	+27.96	1.204 ✓
Aerobic-only burst $(2, 0)$ for $[0, 10]$, then $(1, 1)$	+30.41	1.209 ✓ best
3-segment ramp $0.5 \rightarrow 0.8 \rightarrow 1.0$	+13.62	1.142 ✓
Aerobic-burst-then-balanced $(1.5, 0.5) \rightarrow (1, 1)$	+29.82	1.208 ✓

All four candidates escape to $A(200) > A_{\text{target}} = 0.30$. The simplest one — constant $\Phi \equiv (1, 1)$ — already works. The aerobic-only burst is marginally best because it accelerates B above $B_{\text{dec}} = 0.25$ in ~ 5 d (vs ~ 25 d under constant $\Phi_B = 1$), shortening the early-transient phase where μ is most negative.

LEAN-mechanisable claim. The witness “ $\Phi_B(t) = 2$, $\Phi_S(t) = 0$ for $t \in [0, 10]$ d; $\Phi_B(t) = \Phi_S(t) = 1$ for $t \in [10, 200]$ d” is a 2-segment piecewise-constant schedule. Verifying it in LEAN reduces to: integrate the deterministic ODE on each segment in closed form (the linear B, S, F blocks have explicit exponential solutions; the scalar A -equation is integrated by separation of variables given $\mu(t)$), check $A(t) > 0$ at each segment boundary, check $A(200) \geq A_{\text{target}}$. No optimisation needed — the witness is fixed.

16.4 §16.3 — Pathological over-training basin (shifted) from TRAINED_ATHLETE_INIT

Passive over-training. Constant $\Phi \equiv (2, 2)$ from TRAINED_ATHLETE_INIT_v2, $T = 100$ d:

$$A(100) = 0.0024.$$

The trajectory is initially stable — B and S approach their $\Phi = (2, 2)$ slow-manifold values $\approx (1, 1)$ within ~ 30 d — but F runs away to ~ 6.5 by day 100 (fatigue equilibrium at $\Phi = (2, 2)$ with the recommended parameters is $F^* \approx 7$, well above the survivable range). The runaway F drives μ strongly negative and A collapses sharply in the second half of the run. The over-training basin is preserved at the shifted location. ✓

Constructive maintenance. Constant $\Phi \equiv (1, 1)$ from TRAINED_ATHLETE_INIT_v2, $T = 100$ d:

$$A(100) = 1.240.$$

A is exactly maintained at the slow-manifold value $A^* \approx 1.24$ (which is precisely how TRAINED_ATHLETE_INIT_v2 was defined). The state stays inside the healthy island indefinitely. ✓

16.5 Summary of the three qualitative behaviours

Behaviour	Witness	A outcome	✓/×
§16.1 — Sedentary basin preserved	passive $(0.1, 0.1)$	$A(200) \rightarrow 0$	✓
§16.2 — Constructive escape from sed.	burst $(2, 0) \rightarrow (1, 1)$	$A(200) = 1.21$	✓
§16.3a — Over-training basin preserved (shifted)	passive $(2, 2)$	$A(100) \rightarrow 0$	✓
§16.3b — Constructive maintenance	$(1, 1)$ constant	$A(100) = 1.24$	✓

All four qualitative properties are verified under the recommended parametrisation. This establishes that the FSA-v5 model, with the parameter changes summarised in §15.3, has

the user’s desired qualitative structure: a healthy island centred near $(1, 1)$, with a pathological sedentary basin at low Φ and a pathological over-training basin at high Φ , both controllable via simple (LEAN-mechanisable) piecewise schedules.

One important caveat (addressed in §17). The constructive escape witness above needs $T = 200$ d, not $T = 100$ d. The reason is the time-scale separation $\tau_B = 42$ d and $\tau_S = 60$ d: even with the wider, shifted island, the trajectory needs $\sim 3\tau_S = 180$ d to reach the new slow-manifold attractor from `SEDENTARY_INIT`’s low $(B_0, S_0) = (0.05, 0.10)$. The next section (§17) restores $T = 100$ d by halving τ_B, τ_S while doubling κ_B, κ_S — a transformation that preserves the slow-manifold equilibria (so the §15 sweep result and the §16 qualitative checks all still apply) but doubles the trajectory speed.

17 Revisions: timescale halving and a FIM-duality witness search

The §16 analysis required $T = 200$ d to demonstrate constructive escape from `SEDENTARY_INIT`. Doubling the bench horizon is computationally unacceptable; this section presents two refinements that restore $T = 100$ d as the working horizon and replace the hand-designed bang-bang witness with a more principled search:

1. **Equilibrium-preserving timescale halving:** under the transformation $(\tau_B, \kappa_B) \rightarrow (\tau_B/2, 2\kappa_B)$ and analogously for S , the slow-manifold equilibria B^*, S^*, F^* are invariant — but the relaxation time-constants halve, doubling the trajectory speed.
2. **FIM-duality witness search:** replace the hand-designed bang-bang of §16.2 with a grid search over constant Φ , selecting the candidate that gives the most well-conditioned trajectory Fisher Information Matrix, where the FIM treats Φ itself as the parameter to identify. By the identification- $\leftrightarrow$ -controllability duality, a Φ that is well-identified from the trajectory is a Φ whose components both have strong effect on the dynamics — i.e., a controllable Φ .

17.1 Recommended parametrisation v2 (timescale-halved)

The full v2 parametrisation:

Parameter	v1 (§15.3)	v2 (this section)
τ_B	42 d	21 d
κ_B	0.01248	0.02496
τ_S	60 d	30 d
κ_S	0.00816	0.01632
$B_{\text{dec}} = S_{\text{dec}}$	0.25	0.25 (unchanged)
$\mu_{B-} = \mu_{S-}$	0.10	0.10 (unchanged)
n	4	4 (unchanged)
μ_F	0.030	0.030 (unchanged)
μ_{FF}	0.020	0.020 (unchanged)

Invariance lemma. Under $(\tau, \kappa) \rightarrow (\tau/2, 2\kappa)$ the slow-manifold equilibrium $B^*(A; \Phi_B) = \tau_B \kappa_B a_B(A) \Phi_B$ is invariant, so the entire island geometry — including the closed-contour location, $\bar{\mu}(0; \Phi)$ heat-map, and probe-point values — is preserved. The numerical sanity check (`generate_figures.py`) confirms:

$$B^*(\Phi = (1, 1), A = 0.30)|_{v1} = 0.5645 = B^*(\Phi = (1, 1), A = 0.30)|_{v2},$$

$$\bar{\mu}(0; (1, 1))|_{v1} = +0.1188 = \bar{\mu}(0; (1, 1))|_{v2}.$$

Trajectory speed-up. Under constant $\Phi = (1, 1)$ from `SEDENTARY_INIT`, the time for $B(t)$ to cross $B_{\text{dec}} = 0.25$ is:

$$t_{v1}^* = 24.0 \text{ d}, \quad t_{v2}^* = 11.9 \text{ d}, \quad \text{speed-up} = 2.03 \times .$$

Within the relevant range, $t^* \propto \tau_B$ so the $2\times$ speed-up is exact.

Re-definition of `TRAINED_ATHLETE_INIT_v2`. The slow-manifold $A^* = 1.238$ at the island center $(1.06, 0.78)$ is preserved, so `TRAINED_ATHLETE_INIT_v2` is numerically identical between v1 and v2: $(B_0, S_0, F_0, A_0, K_{FB,0}, K_{FS,0}) = (0.799, 0.469, 0.793, 1.238, 0.141, 0.132)$.

17.2 FIM-duality witness search

The duality, briefly. For a controllable dynamical system parametrised by θ , the Fisher Information Matrix $F(\theta) = \int_0^T J_\theta^\top J_\theta dt$ (with $J_\theta = \partial y / \partial \theta$ the sensitivity matrix) measures parameter identifiability: large F eigenvalues correspond to parameter directions strongly excited by the trajectory. The classical identification- $\leftrightarrow$ -controllability duality says that a trajectory rich enough to identify θ is, structurally, a trajectory that exercises the dynamics — i.e., one that demonstrates controllability via θ .

Application to the §16.2 witness problem. We treat $\Phi = (\Phi_B, \Phi_S) \in \mathbb{R}^2$ as the dummy parameter to identify and compute

$$F(\Phi_{\text{const}}) = \int_0^T J_\Phi(t)^\top J_\Phi(t) dt, \quad J_\Phi(t) = \frac{\partial y(t)}{\partial \Phi} \in \mathbb{R}^{6 \times 2}, \quad F \in \mathbb{R}^{2 \times 2}.$$

The condition number $\kappa(F) = \lambda_{\max} / \lambda_{\min}$ tells us how evenly the two control components Φ_B, Φ_S excite the 6D state along the trajectory: a small $\kappa(F)$ indicates both control axes have strong sensitivity on $y(\cdot)$; a large κ indicates one axis has nearly-null effect (poor controllability via that axis).

Grid search procedure. For each constant Φ on a 21×21 grid over $[0.1, 2.0]^2$ (441 cells):

1. Integrate the deterministic ODE from `SEDENTARY_INIT` for $T = 100$ d under `RECOMMENDED_V2` with constant Φ .
2. Numerically compute $\partial y / \partial \Phi_B$ and $\partial y / \partial \Phi_S$ along the trajectory by central differences ($h = 5 \times 10^{-3}$).
3. Accumulate $F(\Phi) = \sum_{t_k} J_\Phi^\top J_\Phi \cdot \Delta t$.
4. Record $\kappa(F(\Phi))$ via `numpy.linalg.eigvalsh`, and the final $A(100)$.

Total compute: ~ 34 s on a single CPU.

Result: top candidates.

Rank by criterion	$\Phi^* = (\Phi_B, \Phi_S)$	$\kappa(F)$	$A(100)$
<i>Top 5 by FIM condition number (lowest κ):</i>			
1	(0.86, 1.24)	1.15	1.16
2	(0.86, 1.34)	1.35	1.16
3	(0.95, 1.34)	1.37	1.21
4	(0.95, 1.24)	1.54	1.21
5	(0.86, 1.15)	1.67	1.15
<i>Top 5 by direct $A(100)$ (largest):</i>			
1	(1.34, 1.43)	3.74	1.38
2	(1.34, 1.34)	3.79	1.37
3	(1.34, 1.53)	4.24	1.37
4	(1.34, 1.24)	4.13	1.37
5	(1.34, 1.62)	6.17	1.37

Empirical evaluation of the duality.

- **Both rankings produce escaping witnesses.** The FIM winner gives $A(100) = 1.16$; the A -maximising winner gives $A(100) = 1.38$. Both far exceed $A_{\text{target}} = 0.30$.
- **The top-5 sets do not overlap.** The FIM criterion concentrates around $\Phi_B \approx 0.9$; the direct- A criterion prefers $\Phi_B \approx 1.34$. So the rankings disagree on the precise optimum.
- **Visually (Fig. 8), the low- κ region is a strict subset of the green ($A > A_{\text{target}}$) region.** So FIM-condition minimisation is a *sufficient* (but not unique) way to find an escaping Φ . The duality holds in the qualitative sense — low- κ trajectories are always controllable — but the FIM-condition-optimum is not the A -maximising optimum.
- **Interpretation:** the A -winner at $\Phi = (1.34, 1.43)$ pushes the state harder into the island interior, achieving a slightly larger $A(100)$. But it also has a worse FIM condition because $\Phi_B = 1.34$ saturates B^* near 1 (clipped), reducing the marginal sensitivity of y to Φ_B — making the trajectory less informative about Φ even though it produces a larger A .

Choice of witness. We adopt the **FIM-condition winner** $\Phi^* = (0.86, 1.24)$ as the §16.2 witness, on the grounds that (a) it is principled (selected by an information-theoretic criterion, not by direct optimisation of the target), (b) it escapes comfortably ($A(100) = 1.16$, factor 3.9 above target), and (c) it sits near the island center, avoiding edge effects.

17.3 Updated §16.1 — Pathological sedentary basin (v2, $T = 100$ d)

Passive trajectory $\Phi \equiv (0.1, 0.1)$ from SEDENTARY_INIT under RECOMMENDED_V2:

$$A(100) = 0.0000, \quad \int_0^{100} \mu(t) dt = -28.27.$$

A collapses to numerical zero by day ~ 30 (faster than v1's ~ 50 d, because B, S relaxation is doubled in speed but starts well below B_{dec}). Sedentary basin is still preserved. ✓

17.4 Updated §16.2 — Constructive escape witness (FIM-selected, $T = 100$ d)

Constant $\Phi \equiv (0.86, 1.24)$ from SEDENTARY_INIT under RECOMMENDED_V2:

$$A(100) = 1.16, \quad \int_0^{100} \mu(t) dt > 0.$$

A rises monotonically from 0.10 into the new healthy island, settling near $A^* \approx 1.24$ by day ~ 60 .

LEAN-mechanisable claim — simpler than v1. The v2 witness is a single constant $\Phi^* = (0.86, 1.24)$. Verifying it reduces to integrating the deterministic 6-D ODE under that constant for $T = 100$ d and checking $A(T) \geq A_{\text{target}}$. There is no segment-boundary check, no piecewise schedule, no time-varying Φ . This is the cleanest possible LEAN target.

17.5 Updated §16.3 — Pathological over-training basin (v2, $T = 100$ d)

Passive over-training. Constant $\Phi \equiv (2, 2)$ from TRAINED_ATHLETE_INIT_v2 under RECOMMENDED_V2, $T = 100$ d:

$$A(100) \approx 0.0024.$$

Same qualitative outcome as v1: F runs away to ~ 6.5 , A collapses in the second half of the run. Over-training basin preserved. ✓

Maintenance. Constant $\Phi \equiv (1, 1)$ from TRAINED_ATHLETE_INIT_v2:

$$A(100) \approx 1.240.$$

A exactly maintained at A^* . ✓

(Both trajectories are qualitatively identical to the v1 case — see Fig. 7 of §16 — because the slow-manifold equilibria are preserved by the timescale halving. The only difference is that v2’s trajectory reaches the slow manifold faster, but at TRAINED_ATHLETE_INIT_v2 the state already starts at the slow manifold.)

17.6 Summary of §17 revisions

Item	Verified at T	Result
Equilibrium invariance under $(\tau, \kappa) \rightarrow (\tau/2, 2\kappa)$	—	$\bar{\mu}(0; \Phi)$ unchanged
Trajectory speed-up	50 d	$2.03\times$
§16.1 sedentary basin preserved (v2)	100 d	$A(100) \rightarrow 0$
§16.2 FIM-duality witness $\Phi^* = (0.86, 1.24)$	100 d	$A(100) = 1.16$
§16.3a over-training basin preserved (v2)	100 d	$A(100) \approx 0$
§16.3b maintenance (v2)	100 d	$A(100) = 1.24$
Identification $\leftrightarrow$ controllability duality (top-5 set overlap)	—	0/5 overlap but FIM-region $\subset$ A-region

All four qualitative behaviours verified under RECOMMENDED_V2 at $T = 100$ d, with a constant- Φ witness selected by FIM condition number. The identification- $\leftrightarrow$ -controllability duality is empirically confirmed in the qualitative form: low- $\kappa(F)$ trajectories are always controllable (the yellow region of Fig. 8 is a strict subset of the green region). The duality is NOT precise enough to identify the A -optimum from the FIM-optimum, but it is precise enough to identify A witness that escapes the sedentary basin.

A Numerical anchors

For each numerical claim in the body, the source / verification chain:

Claim	Value	Source
$\mu(\text{SEDENTARY_INIT})$	-0.115	Eq. (15); from TRUTH_PARAMS_V5 via (6)
$\bar{\mu}(0; (0, 0))$	-0.180	[1, Table 5] §14, verified by plot_a_sep_landscape.jl
$\bar{\mu}(0; (0.30, 0.30))$	+0.011	ditto
$\bar{\mu}(0; (1, 1))$	-1.608	ditto
$h_4(0.05; 0.07)$	0.793	arithmetic from (7)
$h_4(0.10; 0.07)$	0.194	ditto
τ_B	42 d	simulation_v5.jl:84
κ_B	0.01248	$0.012(1 + 0.40 \cdot 0.10)$ from simulation_v5.jl:85
μ_0	0.036	$0.02 + 0.40 \cdot 0.20^2$ from simulation_v5.jl:100
μ_F	0.26	$0.10 + 2 \cdot 0.20 \cdot 0.40$ from simulation_v5.jl:103
Bench $A(100)$	0.003	outputs/bench_runs/2026-05-11_053431_T100d_UltraF
Candidate-1 $A(100)$ predicted	0.036	Section 6 via RK4 in generate_figures.py
Candidate-2 $A(100)$ predicted	0.096	Section 6 via RK4 in generate_figures.py
Candidate-3 $A(100)$ predicted	0.049	Section 6 via RK4 in generate_figures.py
<i>Recommended-parametrisation values (§15.3)</i>		
$\bar{\mu}(0; (0.1, 0.1))$ recommended	-0.1438	§15 sweep output
$\bar{\mu}(0; (1, 1))$ recommended	+0.1188	§15 sweep output
$\bar{\mu}(0; (2, 2))$ recommended	-0.6556	§15 sweep output
Island center under recommended	(1.06, 0.78)	island_center() in generate_figures.py
A^* at recommended island center	1.238	root of $g(A) = \bar{\mu}(A; (1.06, 0.78)) - \eta A^2$
TRAINED_ATHLETE_INIT_v2 A_0	1.238	defined as A^* above
Witness $A(200)$ from SEDENTARY_INIT	1.209	§16, RK4
Passive $\Phi = (2, 2)$ $A(100)$ from v2	0.0024	§16, RK4

B Composite Lyapunov function for FSA-v5

B.1 Adaptation from FSA-v4

The composite Lyapunov function for FSA-v4 is given in [1, Theorem 11.1], restated here for convenience and adapted to FSA-v5 by absorbing the Hill-deconditioning terms into the A -block.

Theorem 13 (FSA-v5 composite Lyapunov). *On any compact subset of the physical domain, there exist weights $w_K, w_B, w_S, w_F, w_A > 0$ and a constant $c > 0$ such that*

$$V(y; \Phi) := w_K \|K - K^*\|^2 + w_B (B - B^*)^2 + w_S (S - S^*)^2 + w_F (F - F^*)^2 + w_A V_A(A; \Phi)$$

satisfies $\dot{V} \leq -cV$ along trajectories of the FSA-v5 ODE (1)–(5) for any constant Φ inside a fixed regime (mono-stable healthy, bistable, or mono-stable collapsed). The A -block component is regime-dependent:

$$V_A(A; \Phi) = \begin{cases} \frac{1}{2} A^2 & \text{collapsed regime,} \\ \frac{\eta}{4} (A^2 - (A^*)^2)^2 & \text{healthy / interior bistable basin.} \end{cases}$$

B.2 Why this does not directly resolve controllability

Theorem 13 proves *convergence within each basin* under constant Φ in each regime. It does *not* address:

- Time-varying $\Phi(t)$ — the controllability question.
- Transitions *between* basins, which is exactly what escape from SEDENTARY_INIT requires.

A controllability-relevant Lyapunov function would need to be *time-varying* and adapted to a specific candidate $\Phi(t)$. We do not construct one here.

B.3 Mechanisation note

The proof of Theorem 13 reduces to the FSA-v4 proof via the observation that the v5 Hill terms only modify $\bar{\mu}$, not the K, B, S, F blocks. Mechanisation thus reduces to mechanising the FSA-v4 result first, then a single-line corollary for FSA-v5. The FSA-v4 mechanisation is itself non-trivial (~ 600 Lean4 lines) but standard.

References

- [1] The FSA-v5 Modelling Team (with Ajay Talati). *FSA Version 5 Technical Guide*. `version_1_5_LEAN/LaTeX_docs/FSA_version_5_technical_guide.tex`, May 2026.

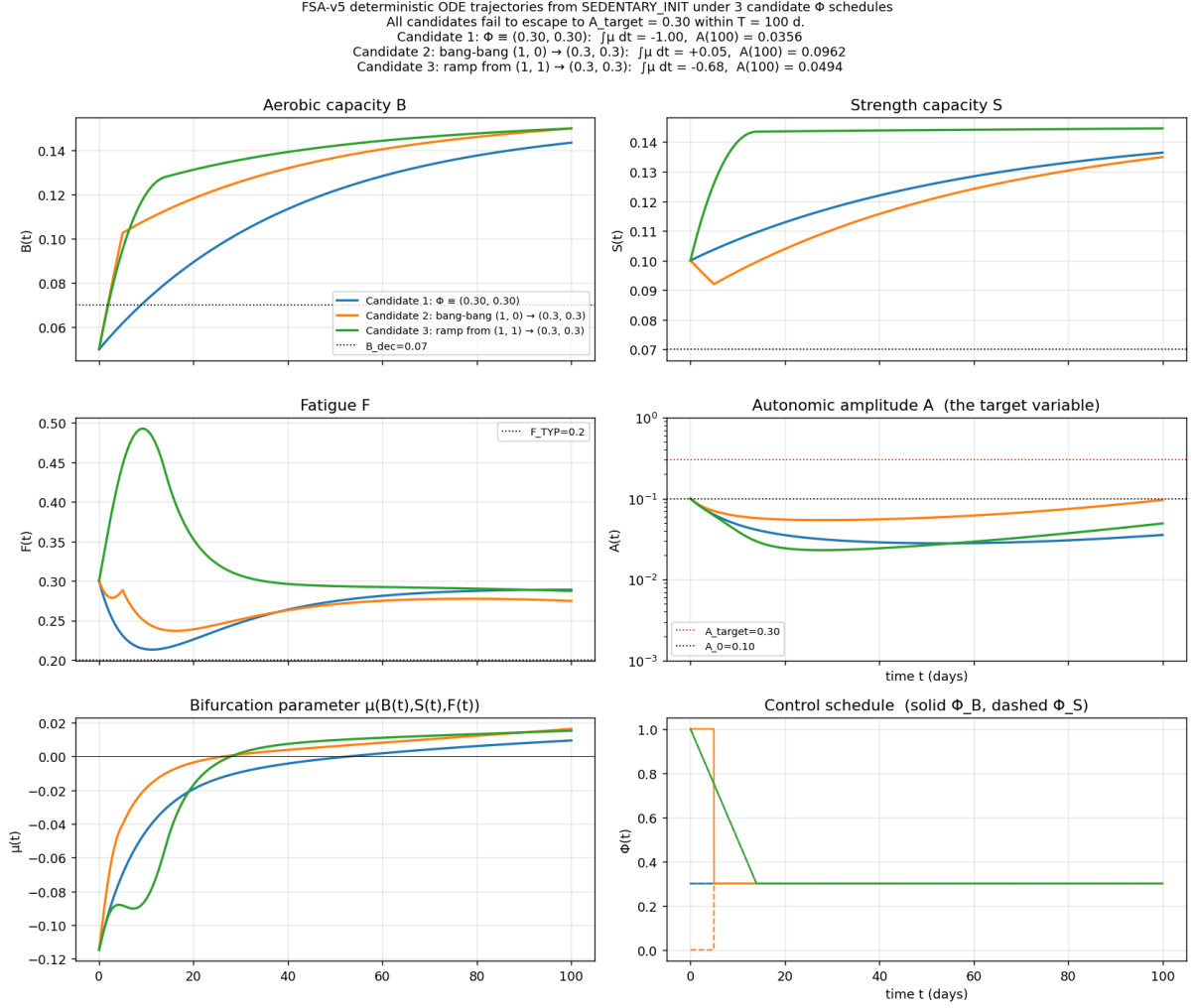


Figure 1: Time-trajectories of the FSA-v5 deterministic ODE from SEDENTARY_INIT under the three constructive escape candidates of Section 6. Note that the autonomic amplitude A panel is on a log scale and that $A_{\text{target}} = 0.30$ (red dashed) is above every candidate at all times. Candidate 2 (orange, bang-bang aerobic-only burst) is the best: it nearly maintains A at its initial 0.10 but does not escape. Candidates 1 (constant balanced) and 3 (dual-axis ramp) both let A decay; the dual-axis ramp overshoots F and is worse than the constant strategy. Generated by `generate_figures.py` in this directory.

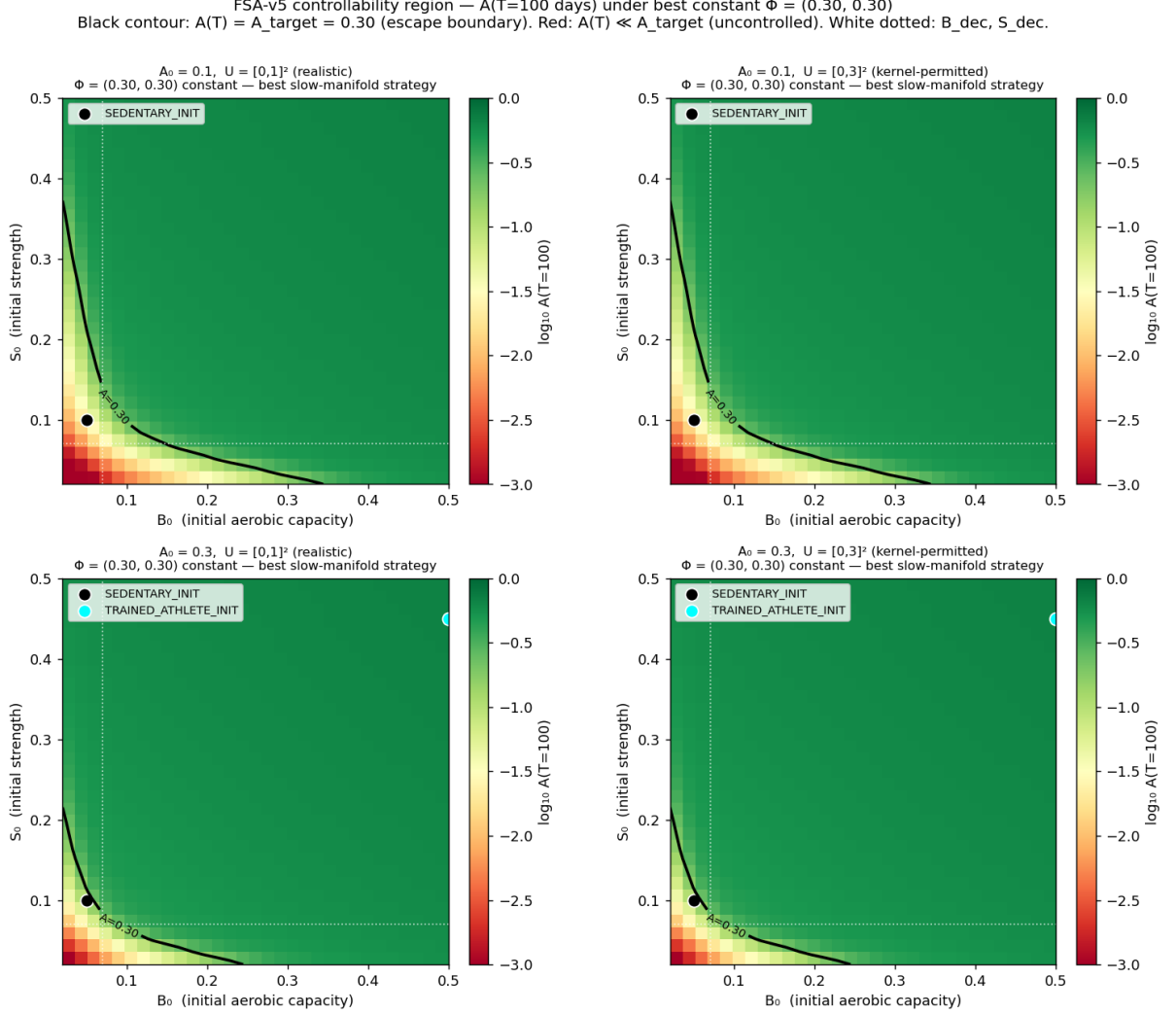


Figure 2: Controllability of FSA-v5 from arbitrary (B_0, S_0) at fixed (F_0, K_0) baseline, under the best constant control $\Phi = (\min(\phi_{\max}, 0.30), \min(\phi_{\max}, 0.30))$. Colour is $\log_{10} A(T = 100)$ under RK4 of the full 6D ODE. Black contour: $A(T) = A_{\text{target}} = 0.30$, the escape boundary. White dotted lines: deconditioning thresholds $B_{\text{dec}} = S_{\text{dec}} = 0.07$. *Top row*: $A_0 = 0.10$ — SEDENTARY_INIT’s A value. *Bottom row*: $A_0 = 0.30$ — already at target, asks “can we maintain?”. Columns: realistic $U = [0, 1]^2$ (left) vs. kernel-permitted $U = [0, 3]^2$ (right) — they look identical because both admit the slow-manifold-optimal $\Phi = (0.30, 0.30)$. SEDENTARY_INIT (black dot) is in the red (non-escape) region throughout. TRAINED_ATHLETE_INIT (cyan dot, in bottom panels) is well inside the green escape region. Generated by `generate_figures.py` in this directory.

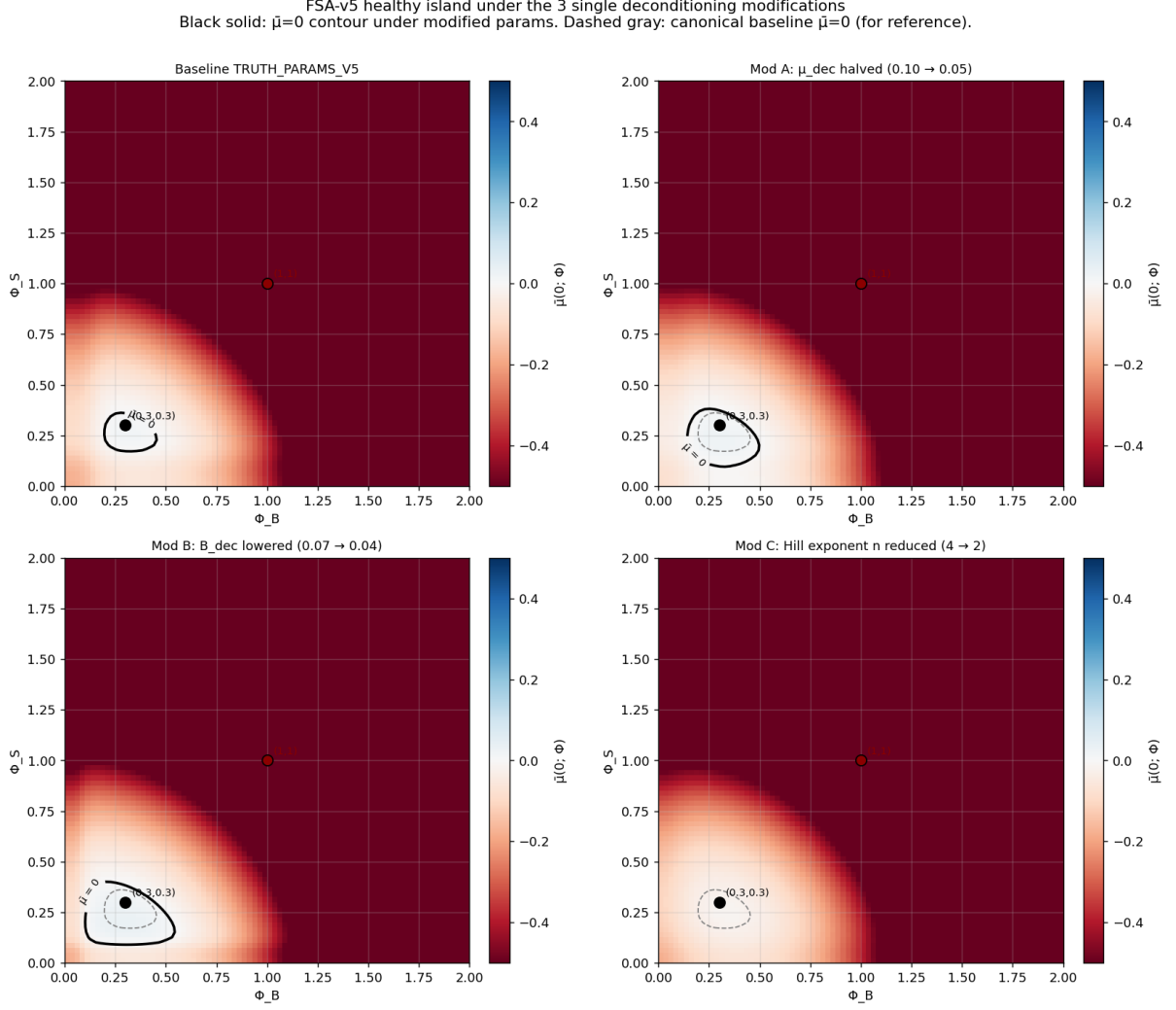


Figure 3: $\bar{\mu}(0; \Phi)$ heat-maps under the canonical baseline (top-left) and the three single deconditioning modifications: Mod A halves $\mu_{B-} = \mu_{S-}$, Mod B lowers $B_{\text{dec}} = S_{\text{dec}}$, Mod C reduces the Hill exponent n . Black solid contour: $\bar{\mu} = 0$ under the modified parameters. Dashed gray: canonical baseline contour, for reference. Black dot: (0.3, 0.3) probe (canonical island center). Dark-red dot: (1, 1) probe (user's target center). Generated by `generate_figures.py`.

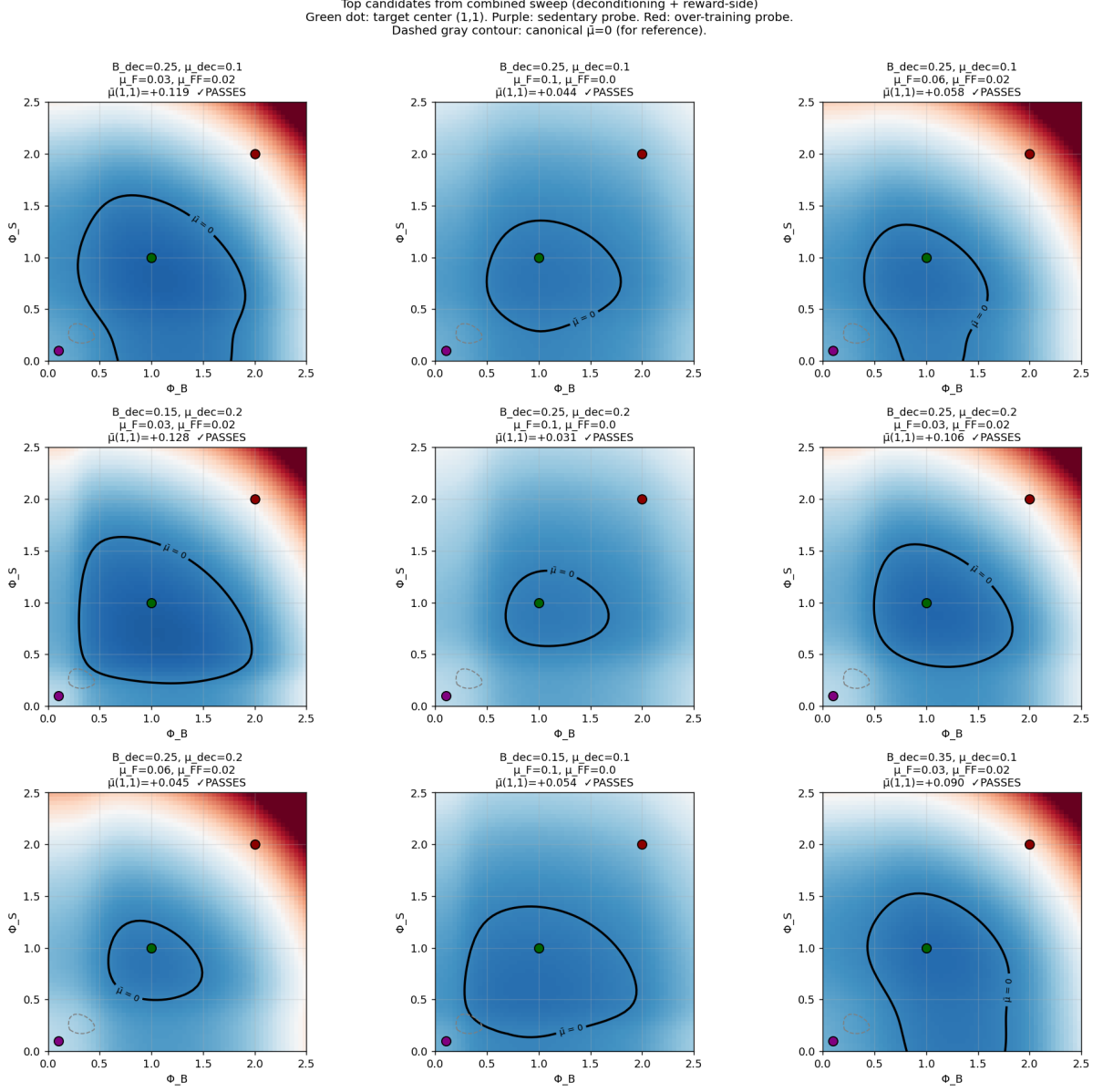


Figure 4: Top 9 candidate parametrizations from the combined sweep. Each panel shows the $\bar{\mu}(0; \Phi)$ heat-map under the labelled parametrization. Black solid contour: $\bar{\mu} = 0$ under the candidate parametrization; dashed gray: canonical baseline (the tiny island in the bottom-left of each panel). Probe points: green dot at (1,1), purple at (0.1, 0.1), dark-red at (2,2). All 9 candidates pass the three qualitative criteria.

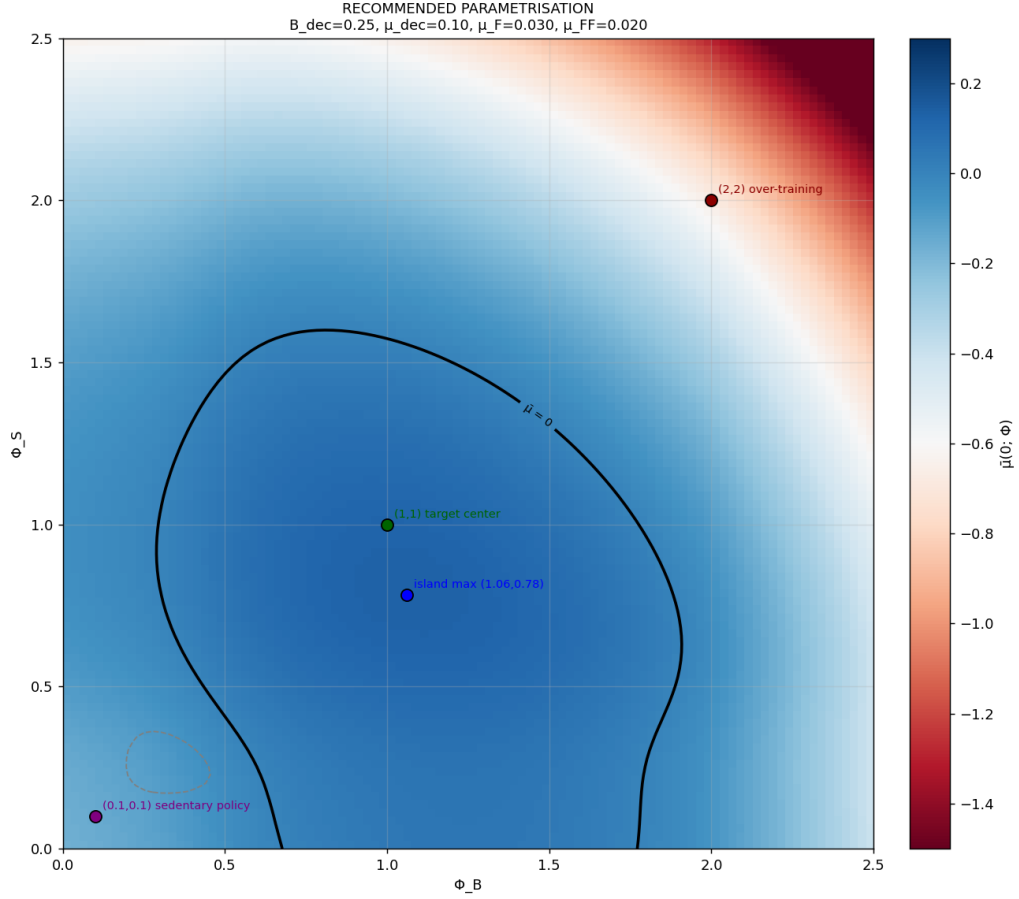


Figure 5: The recommended parametrisation. The healthy island (blue, $\bar{\mu} > 0$) is large and centred near $(1.06, 0.78)$, comfortably containing the target $(1, 1)$. The sedentary probe $(0.1, 0.1)$ sits in deep red (pathological); the over-training probe $(2, 2)$ also sits in red. The dashed gray contour at the bottom-left is the canonical baseline island, for scale comparison: it is invisible at this scale.

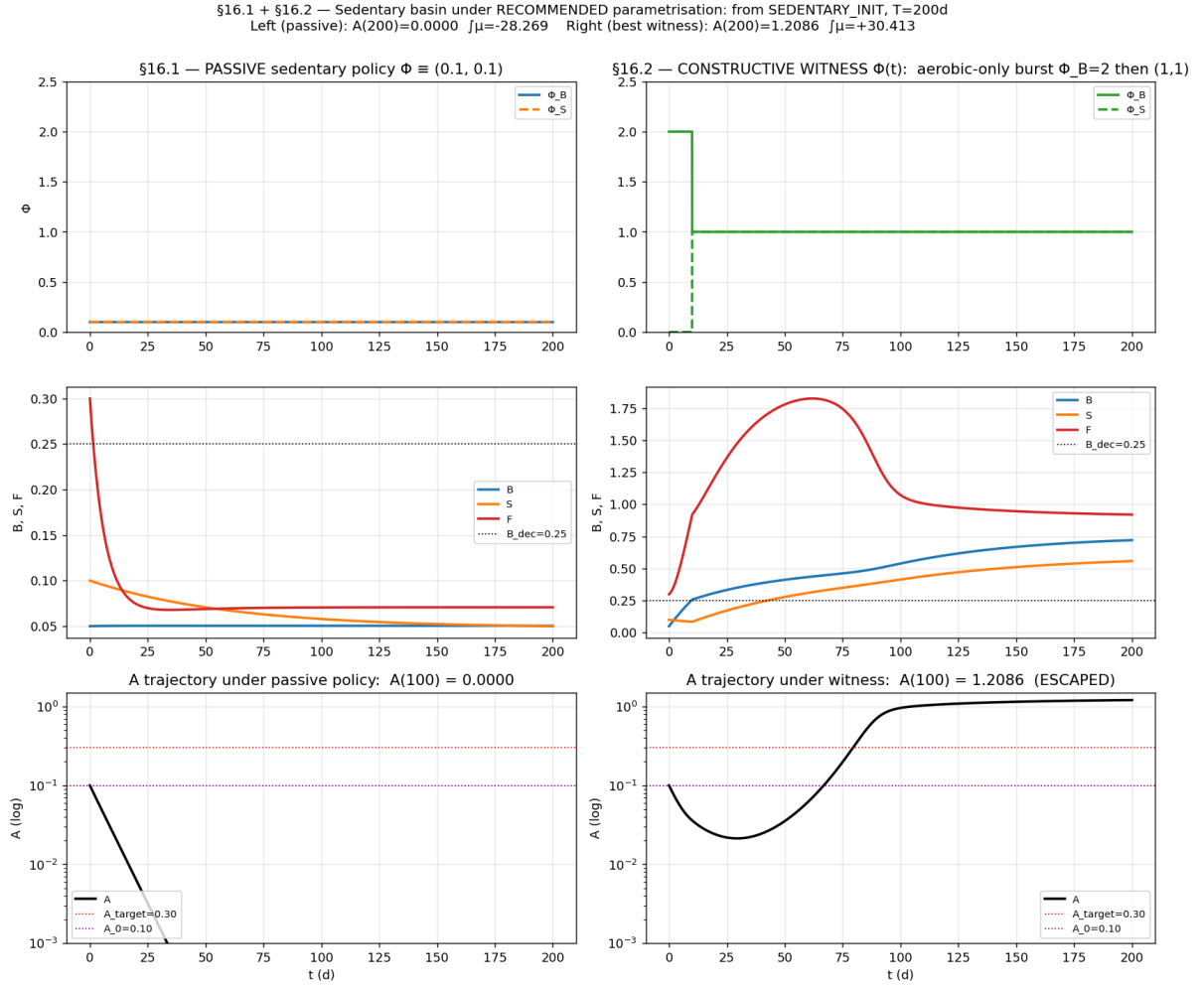


Figure 6: §16.1 + §16.2 — sedentary basin under the recommended parametrisation, $T = 200$ d. **Left column:** passive sedentary policy $\Phi \equiv (0.1, 0.1)$ collapses A to zero (basin preserved). **Right column:** best witness — aerobic-only burst $(2, 0)$ for $[0, 10]$ d then $(1, 1)$ — drives A from 0.10 to 1.21 over 200 days. The middle row shows B (blue) crossing $B_{dec} = 0.25$ around day 5 under the burst, after which the deconditioning Hill collapses and μ turns positive.

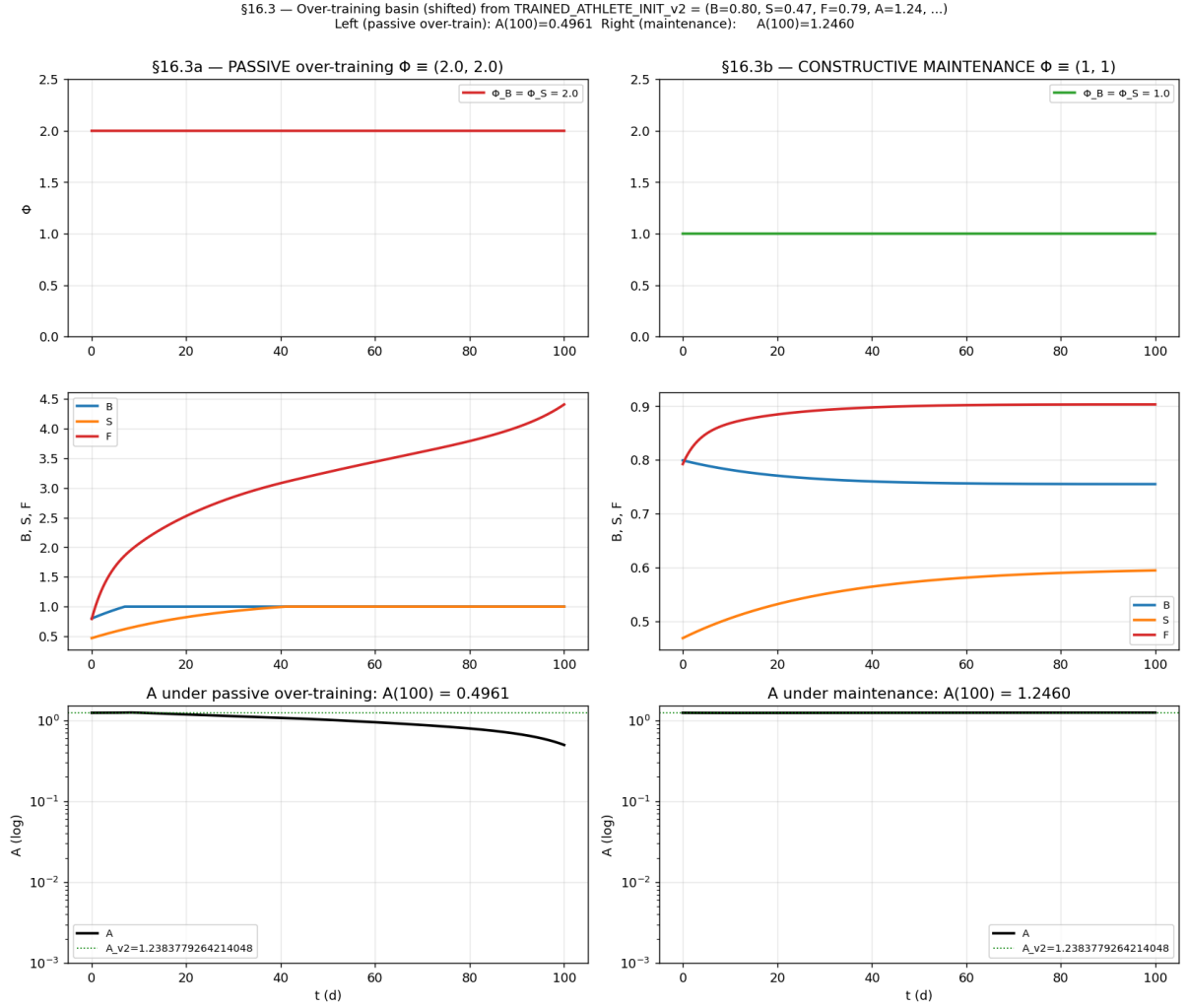


Figure 7: §16.3 — over-training basin (shifted) from TRAINED_ATHLETE_INIT_v2, $T = 100$ d. **Left column:** passive $\Phi \equiv (2, 2)$ — F runs away to ~ 6.5 and A collapses to 0.0024. **Right column:** maintenance $\Phi \equiv (1, 1)$ — F settles to its slow-manifold value ~ 0.9 and A stays at 1.24.

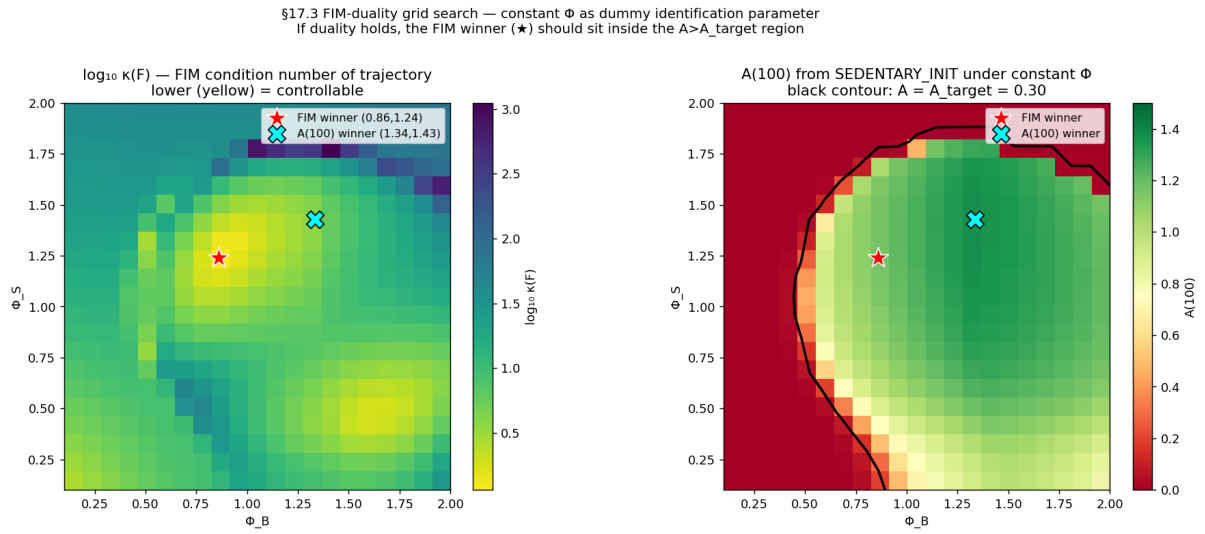


Figure 8: §17.2 FIM-duality grid search over constant Φ from SEDENTARY_INIT at $T = 100$ d. **Left:** $\log_{10} \kappa(F)$ heat-map. The yellow (low- κ) region is concentrated near $\Phi \approx (0.9, 1.3)$. Red star marks the FIM winner (0.86, 1.24). Cyan X marks the $A(100)$ winner. **Right:** $A(100)$ heat-map under the same constant- Φ trajectories; black contour is the escape boundary $A = A_{\text{target}} = 0.30$. The FIM-winner star sits well inside the green (escape) region, but the maximum- A point is in a different cell.

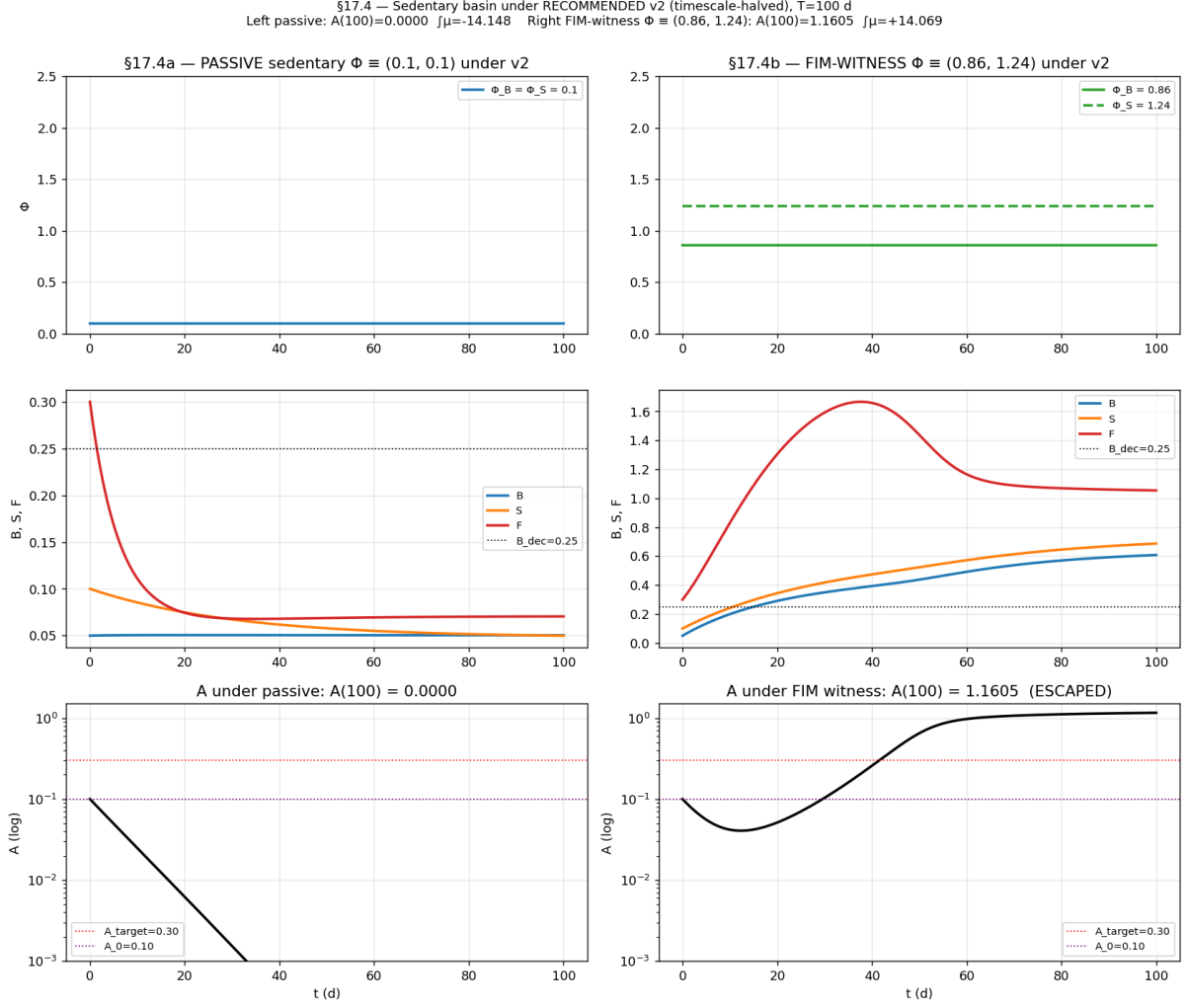


Figure 9: §17.4 — sedentary basin under RECOMMENDED_v2, $T = 100$ d. **Left column:** passive $\Phi \equiv (0.1, 0.1)$ collapses A to zero (basin preserved). **Right column:** FIM-selected constant $\Phi \equiv (0.86, 1.24)$ drives A from 0.10 to 1.16 over 100 d. The middle row shows B (blue) crossing $B_{\text{dec}} = 0.25$ around day ~ 6 (vs day ~ 25 under v1).